

TARGET-90

JKBOSE CLASS-11TH

STRUCTURE OF ATOM

One shot

CHEMISTRY

Vishal Sir



Bhaichara
ON
TOP



JKBose Wallah

Topics To Be Covered



1 ... Structure of Atom

2 ...

3 ...

4 ...

JKBOSE Syllabus.

5802

Hydrocarbons

5 Marks



Unit-II: STRUCTURE OF ATOM

09 Marks

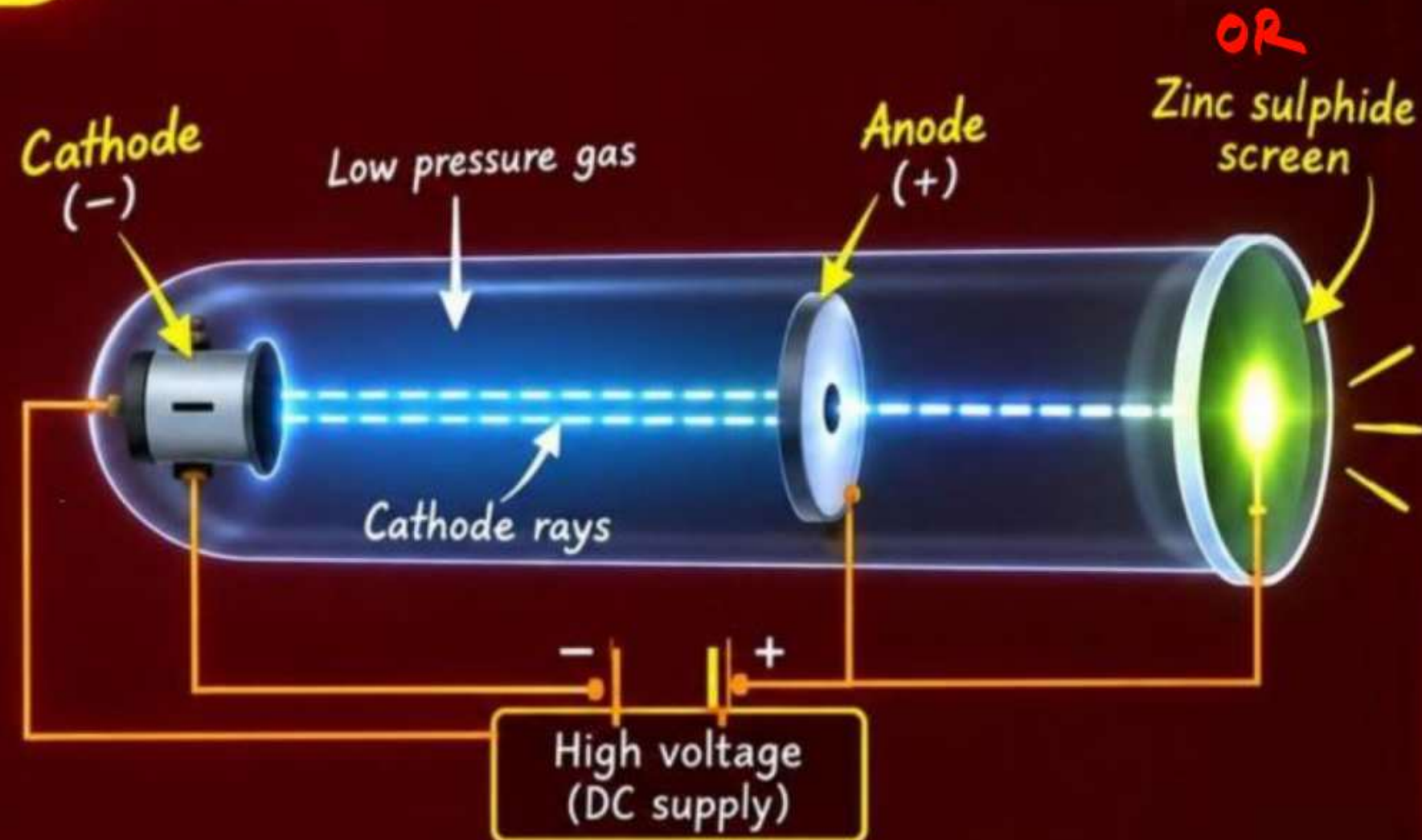
Discovery of electron, proton and neutron, atomic number, isotopes and isobars. Thompson's model and its limitations, Rutherford's model and its limitations. Bohr's model & its limitations, concept of shells and sub-shells. Dual nature of matter and light, de-Broglie's relationship. Heisenberg's uncertainty principle, concept of orbitals, quantum numbers, shapes of s, p and d- orbitals. Rules for filling electrons in orbitals- Aufbau's principle, Pauli's exclusion principle and Hund's rule. Electronic configuration of atoms, stability of half-filled and completely filled orbitals.

Discovery of Electron



Fluorescence

- **J.J. Thomson** (late 19th century) discovered the electron.
- Used a **discharge tube experiment** (glass tube with **low pressure gas** and **high voltage**).
- **Cathode rays** (stream of particles) move from **cathode (-)** to **anode (+)**.
- A small hole was made in the anode and a **zinc sulphide screen** was placed behind it.
- A **bright spot** on the screen confirmed that cathode rays travel in a **straight line**.



Bright spot on the screen

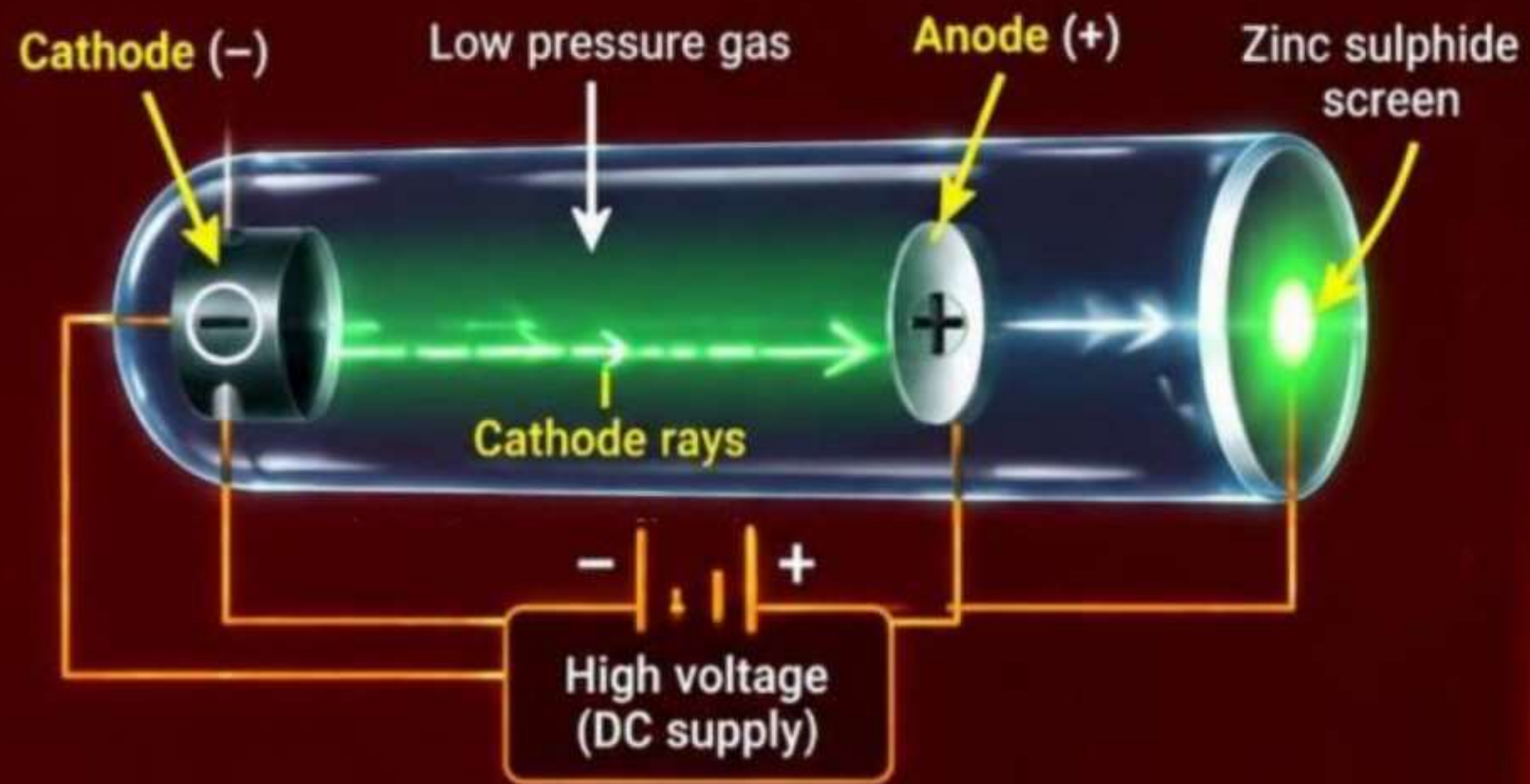
→ Confirms path of cathode rays

→ Electrons move in a straight line

Properties of Cathode Rays



-  They travel in **straight lines**.
-  They are made of **material particles** (electrons).
-  They carry **negative charge**.
-  They produce a **heating effect** when they strike a surface.
-  They cause **ionization** of gases through which they pass.
-  Their properties **do not depend** on the material of electrodes or the gas in the tube.

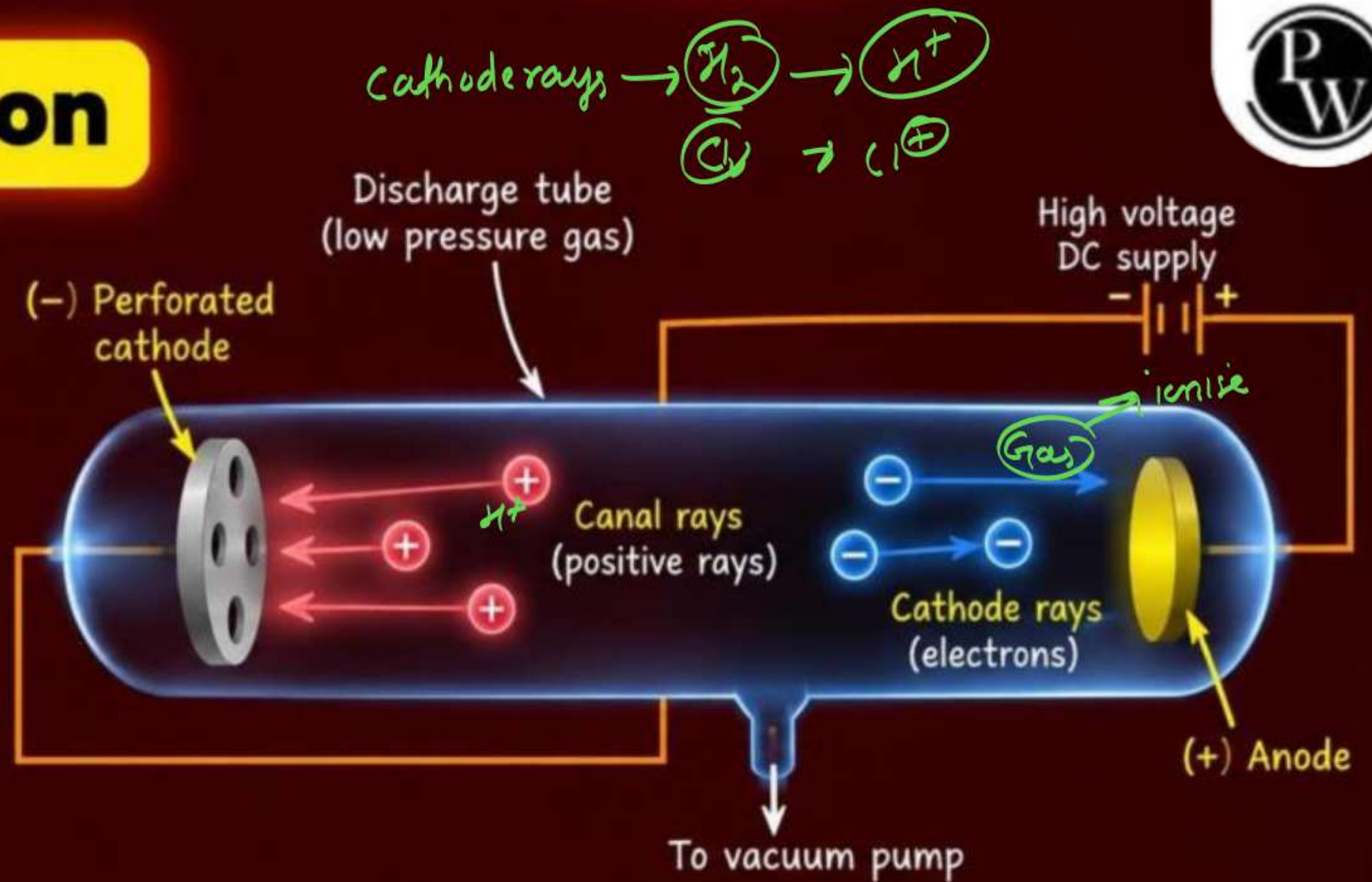


Key Takeaway

Cathode rays are a stream of **negatively charged particles** (electrons) that move in **straight lines** and have the same properties regardless of the electrodes or gas used.

Discovery of Proton

- In 1836, E. Goldstein found a new kind of rays in a discharge tube (in addition to cathode rays).
- These rays were found **behind a perforated cathode** and travelled in the **opposite direction** to cathode rays (i.e., **towards the cathode**).
- These rays consist of **positively charged particles**.
- They were named **positive rays**, or **canal rays**, or **anode rays**.



Key Takeaway

Canal rays are a stream of **positively charged particles** (protons) that move **towards the cathode**, **opposite** to cathode rays.

Characteristics of Anode Rays

$$\left(\frac{e}{m}\right)_p < \left(\frac{e}{m}\right)_e$$

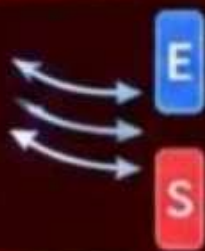


1



They move in **straight lines** and form **shadows** of objects in their path.

2



They get **deflected by electric and magnetic fields**, but in the **opposite direction** to cathode rays (showing they are positively charged).

3



They can **rotate a light wheel** (material particle) and also produce a **heating effect**.

4

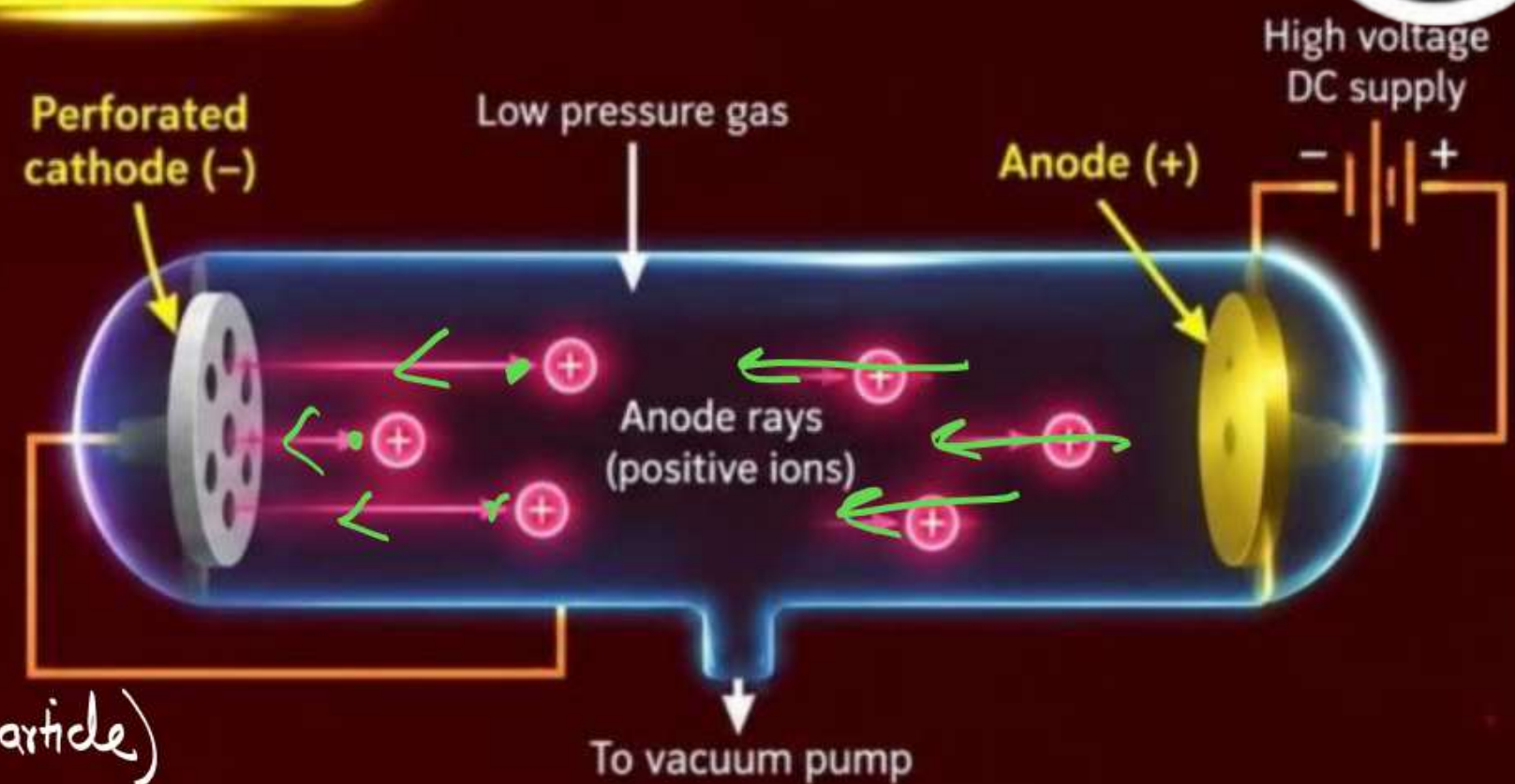
$$\frac{e}{m}$$

Their charge/mass ratio (e/m) is **much smaller** than electrons.

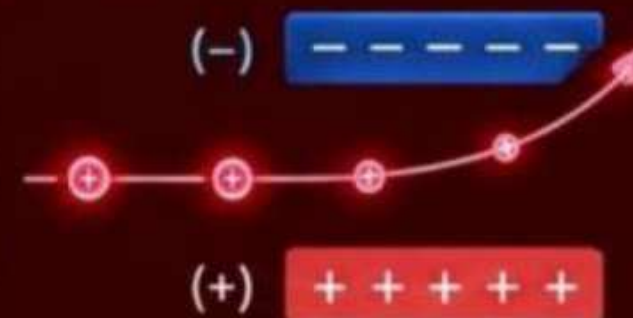
5



The e/m value depends on the **type of gas** present in the tube.

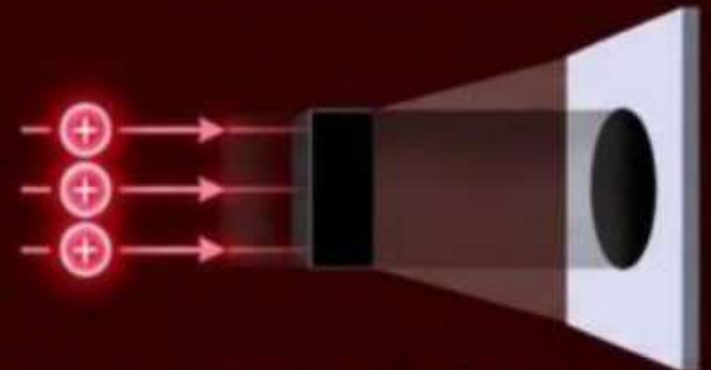


Deflection in Electric Field



Deflected towards the negative plate (opposite to cathode rays).

Shadow Formation



Move in straight lines and form shadows.

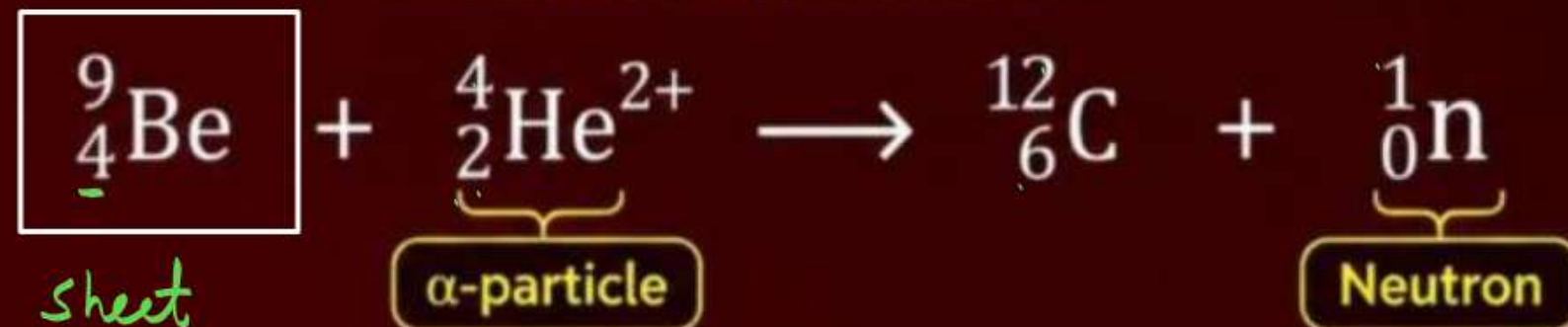
- Cathode rays originate from cathode. (electron)
- Anode rays originate from gas present in a discharge tube.

When there is H_2 gas present in discharge
then the particle of anode rays is called proton (H^+).

Discovery of Neutron

- In **1932**, **James Chadwick** bombarded a thin **beryllium (Be)** sheet with **α -particles** (He^{2+}).
- He observed **highly penetrating** rays which consisted of **neutral particles**.
- These neutral particles were found to have a mass of **$1.675 \times 10^{-27} \text{ kg}$** , which is nearly the same as that of a hydrogen atom, and they carry **no charge**.
- These particles were named **neutrons**.

Nuclear reaction



Neutron (n)

✓ **Mass** = $1.675 \times 10^{-27} \text{ kg}$ (1)
($\approx$ mass of hydrogen atom)

✓ **Charge** = 0 (neutral)

✓ **Symbol** = n

Thus, a **neutron** is a sub-atomic particle having **mass equal to that of hydrogen atom** and carrying **no electrical charge**.

Electron (e^-)

Thus, an electron may be defined as a sub-atomic particle which carries **one unit negative charge** ($1.6022 \times 10^{-19} \text{ C}$) and has a mass ($9.1 \times 10^{-31} \text{ kg}$) equal to $1/1837^{\text{th}}$ of that of hydrogen atom.



Electron (e^-)

$$\text{Charge} = -1.6022 \times 10^{-19} \text{ C}$$

$$\text{Mass} = 9.1 \times 10^{-31} \text{ kg}$$

$$= 1/1837^{\text{th}} \text{ of H atom}$$

Proton (p^+)

Thus, a proton is a sub-atomic particle which carries **one unit positive charge** ($1.6022 \times 10^{-19} \text{ C}$) and has mass ($1.67 \times 10^{-27} \text{ kg}$) equal to that of an atom of hydrogen.



Proton (p^+)

$$\text{Charge} = +1.6022 \times 10^{-19} \text{ C}$$

$$\text{Mass} = 1.67 \times 10^{-27} \text{ kg}$$

$$= \text{equal to H atom}$$



$$(e/m)_p < (e/m)_e$$

Specific charge
or
charge/mass ratio

Particle	Charge (C)	Unit Charge	e/m (C kg ⁻¹)	Mass (kg)
 Electron (e ⁻)	1.6×10^{-19} C	-1	<u>1.75×10^{11}</u>	9.1×10^{-31} kg
 Proton (p ⁺)	1.6×10^{-19} C	+1	<u>9.5×10^7</u>	1.67×10^{-27} kg (≈ 1)
 Neutron (n)	0	0	<u>0</u>	1.67×10^{-27} kg (≈ 1)

Thompson Model (HW)

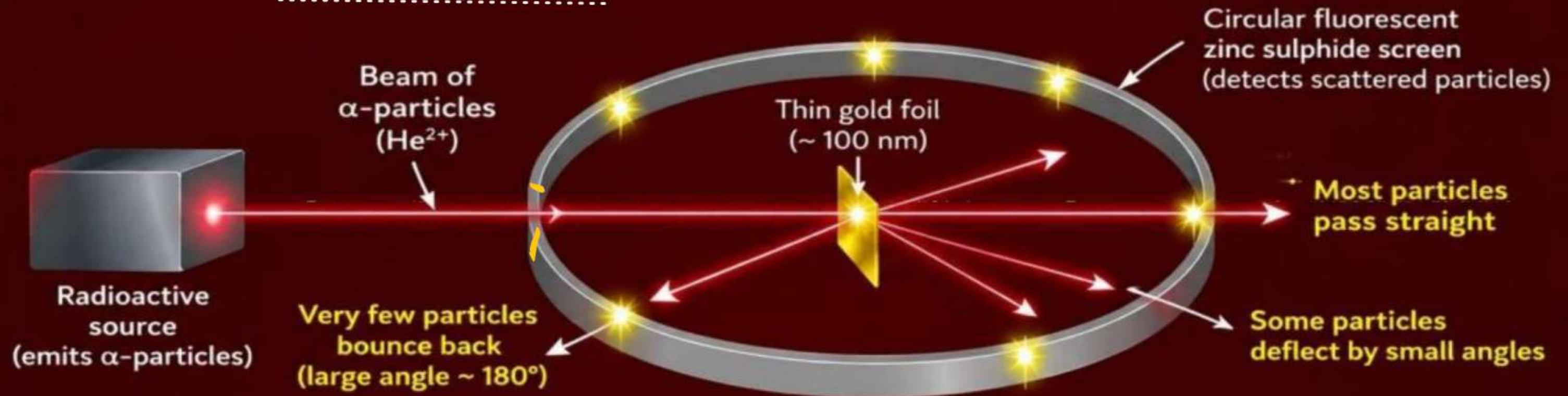
RUTHERFORD MODEL OF ATOM

(Discovery of Nucleus)



Experiment:

- A beam of high energy **α -particles (He^{2+})** from a radioactive source was directed at a very thin **gold foil** (thickness $\sim 100 \text{ nm}$).
- A circular fluorescent **zinc sulphide screen** was placed around the foil to detect the scattered α -particles.



Observations:

- | | | |
|--|---|---|
| 1 Most α -particles passed through the foil undeflected . | 2 A small fraction deflected by small angles . | 3 A very few particles (~ 1 in 20,000) bounced back (nearly 180°). |
|--|---|---|

Conclusions of Rutherford's Experiment

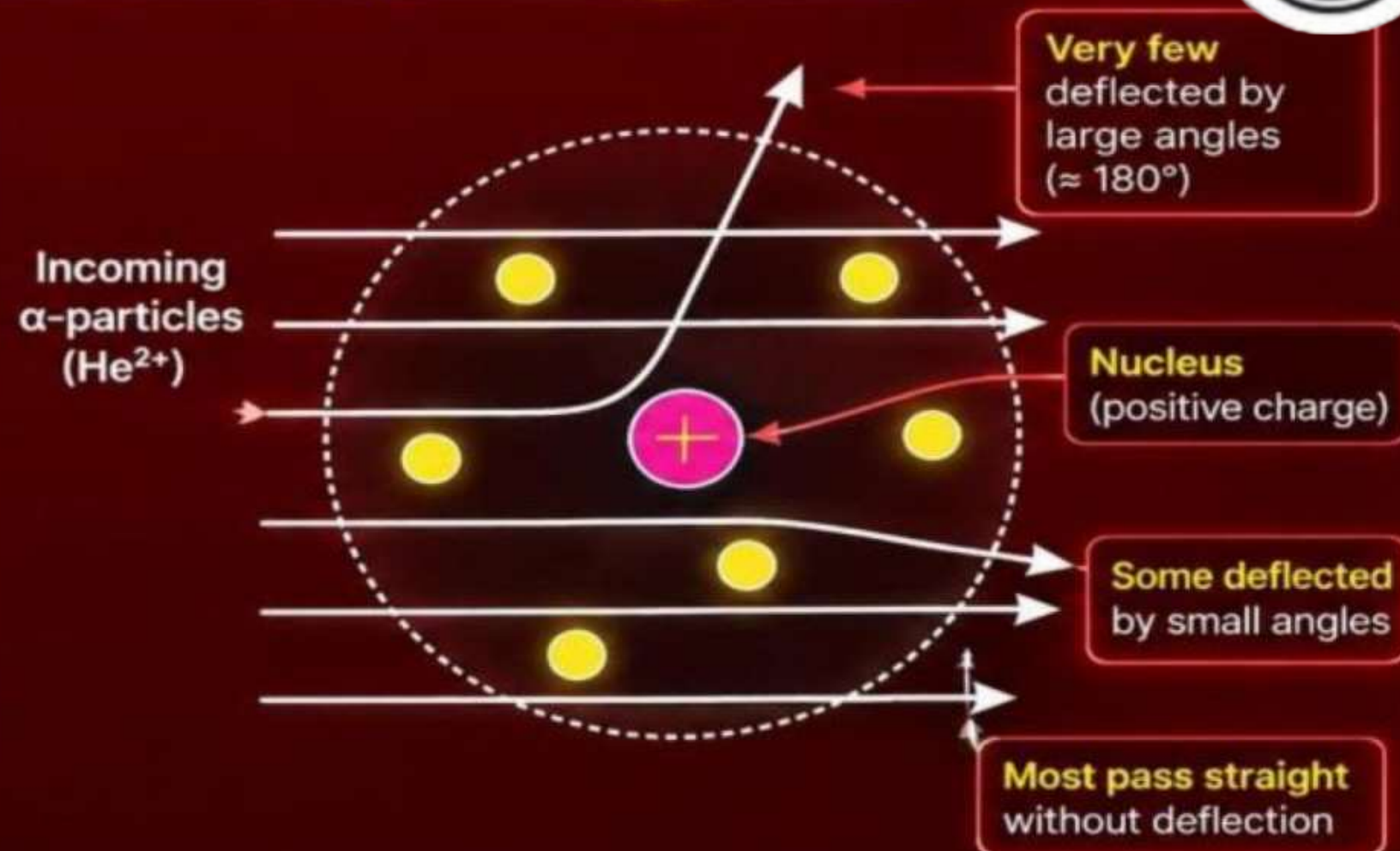


1 Most of the atom is **empty space** because most of the α -particles passed through the gold foil without any deflection.

2 Some α -particles were **deflected**, which means there is a small, **positively charged region** (nucleus) that repels them.

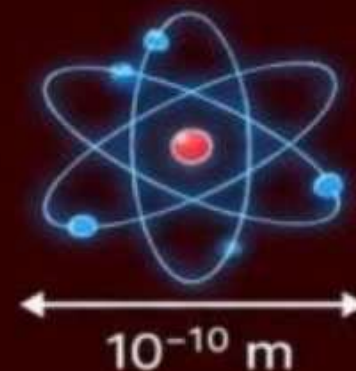
3 The nucleus is **very small** compared to the atom.

- Radius of atom $\approx 10^{-10} \text{ m}$
- Radius of nucleus $\approx 10^{-15} \text{ m}$

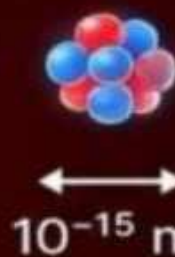


Size comparison (not to scale)

Atom radius
 $\approx 10^{-10} \text{ m}$



Nucleus radius
 $\approx 10^{-15} \text{ m}$

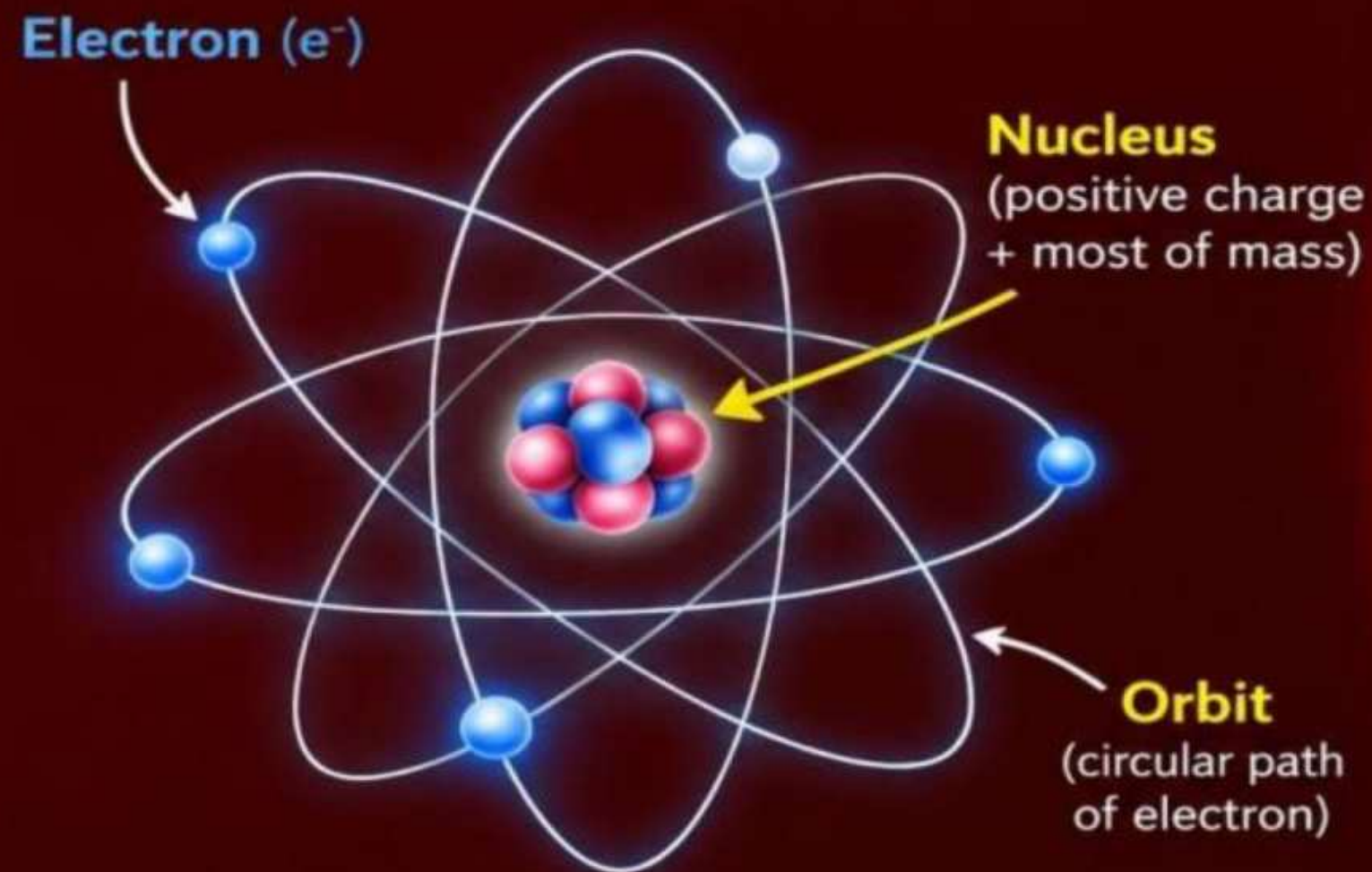


RUTHERFORD MODEL OF ATOM



On the basis of the observations and conclusions of Rutherford's experiment, he proposed the **nuclear model of atom**.

- 1 Positive charge and most of the mass of the atom are concentrated in a very small region called **nucleus**.
- 2 Electrons revolve around the nucleus in circular paths called **orbits** with very high speed.
- 3 Electrons and the nucleus are held together by electrostatic **forces of attraction**.



So, according to Rutherford, an atom has a **tiny, positively charged nucleus** at the centre and **electrons revolve around it in orbits**, held by **electrostatic attraction**.

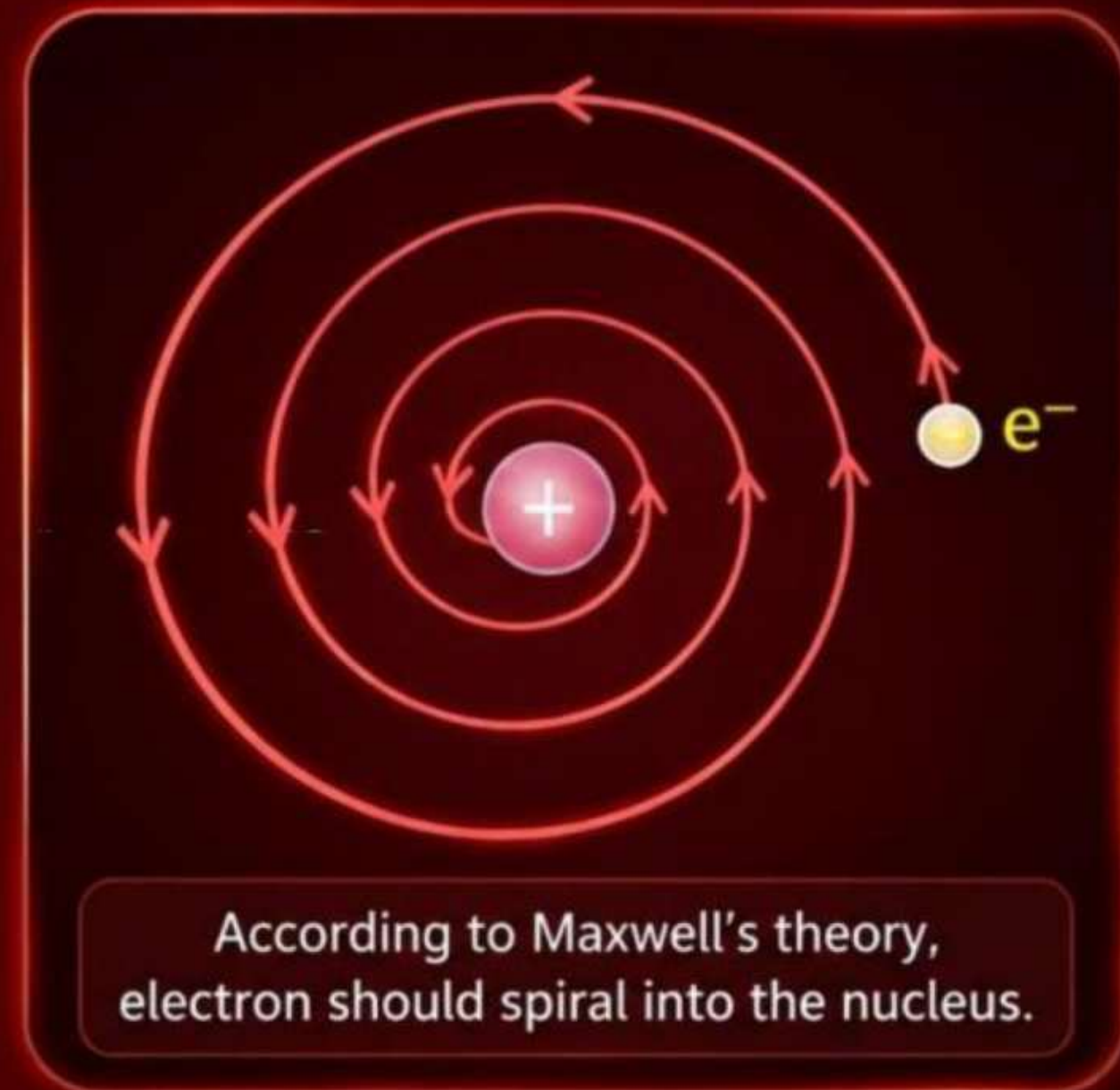
RUTHERFORD MODEL OF ATOM : LIMITATIONS

1 Atomic Stability

- According to **Maxwell's theory**, a moving electron continuously **loses energy**.
- So, it should **spiral into the nucleus** and the atom should collapse in about **10^{-8} seconds**.
- But atoms are stable in reality, so Rutherford's model could not explain **atomic stability**.

2 Distribution of Electrons

- The model could not explain how electrons are **arranged around the nucleus** or why they have **different energies**.



Atomic Number & Mass Number

• Atomic Number (Z):

Number of protons in an atom.

$$Z = \text{No. of } p.$$

- In a neutral atom: **protons = electrons**.

• Mass Number (A):

Number of protons + neutrons.

$$A = p + n$$

- Electrons have almost no mass, so the mass of an atom mainly comes from **protons and neutrons** (nucleons).

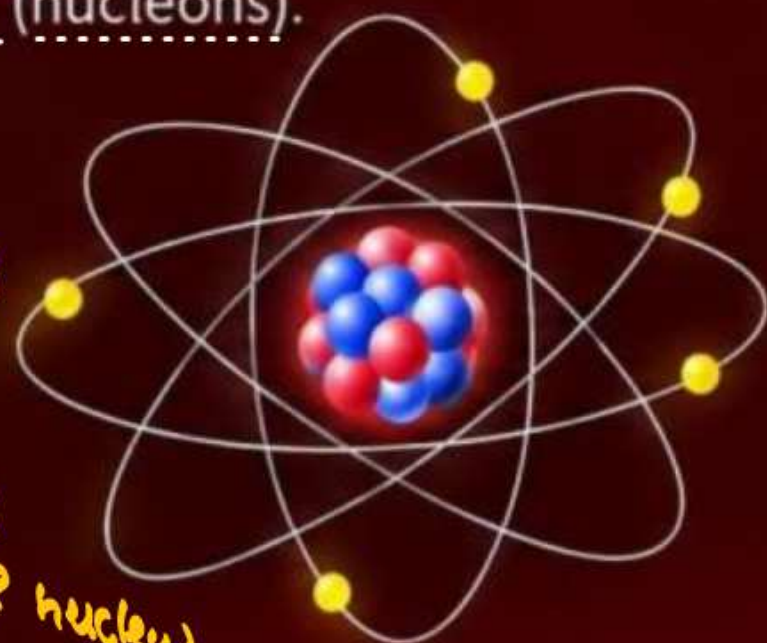
 = Proton (p)

 = Neutron (n)

 = Electron (e⁻)

→ Nucleus

→ Outside the nucleus.



Isotopes, Isobars and Isotones



Isotopes

- Same element (same atomic number), but different mass numbers.
- **Example:** ${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$ (Hydrogen isotopes)

Z same
A different



Isobars

- Different elements with same mass number, but different atomic numbers.
- **Example:** ${}^{40}_{18}\text{Ar}$, ${}^{40}_{19}\text{K}$, ${}^{40}_{20}\text{Ca}$

A same
Z different



Isotones

- Different elements with same number of neutrons.
- **Example:** ${}^{14}_6\text{C}$, ${}^{15}_7\text{N}$, ${}^{16}_8\text{O}$ (all have 8 neutrons)

N same
Z different



Note: $A = Z + N \Rightarrow N = A - Z$

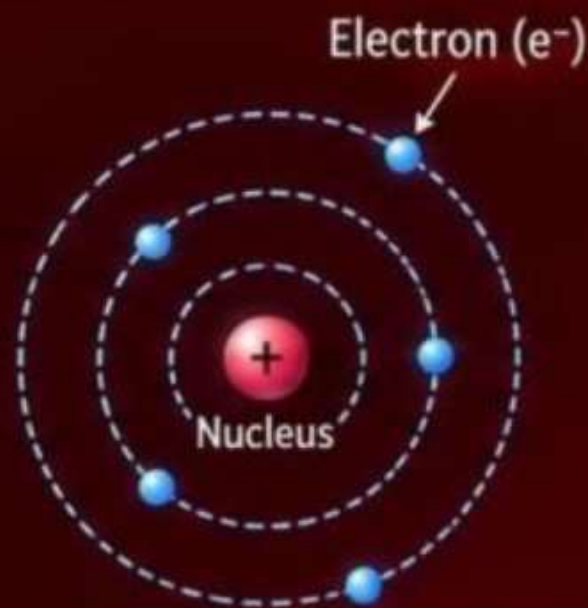
Bohr's Model of Atom (1913)

Valid for
(One electron species)



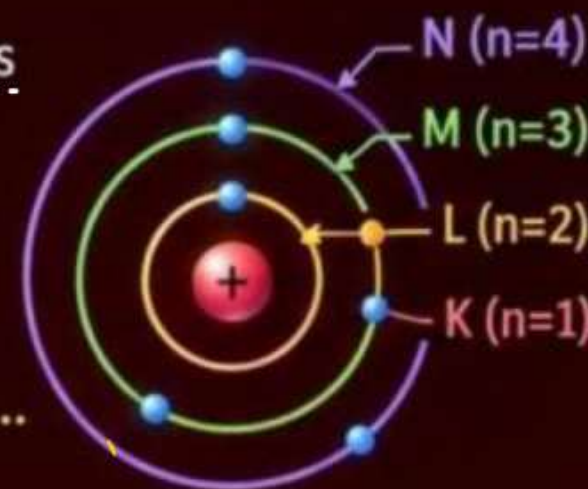
1 Structure

- Atom has a small, heavy, positively charged **nucleus** at the center.
- Electrons move around it in **circular orbits**.



2 Energy Levels (Shells)

- Electrons can revolve only in certain orbits with **fixed energy**.
- These orbits are called **energy levels** (or stationary states).
- Labeled as **K, L, M, N ...** or numbers **1, 2, 3 ...** (closest to nucleus = lowest energy).



3 Quantized Energy

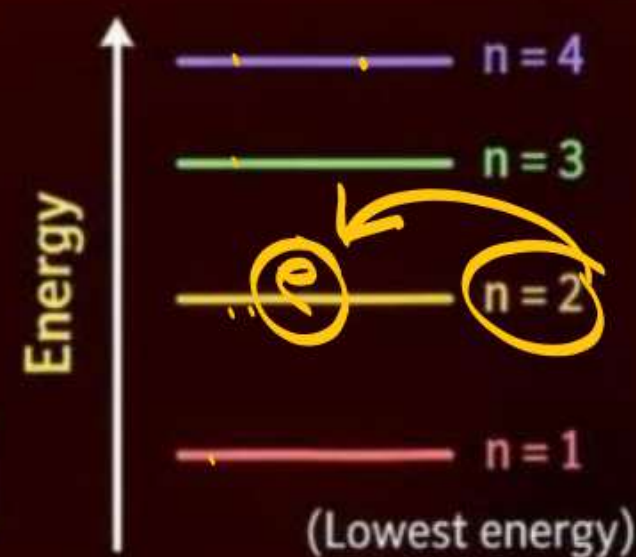
- Electrons can have only specific (**discrete**) energy values, not any random energy.
- For **hydrogen-like atoms**, energy of nth level:

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV atom}^{-1}$$

$$E_n = -\frac{1312 Z^2}{n^2} \text{ kJ mol}^{-1}$$

Order of energy:

$$E_1 < E_2 < E_3 < E_4 < \dots$$



Q. Calculate the energy of 5th shell. for H-atom (He^+)

Sol. $E_5 = -\frac{13.6 Z^2}{n^2} = -\frac{13.6 \times 1^2}{5^2} = -\frac{13.6}{25} \text{ eV/atom.}$

Bohr's Model of Atom (1913)

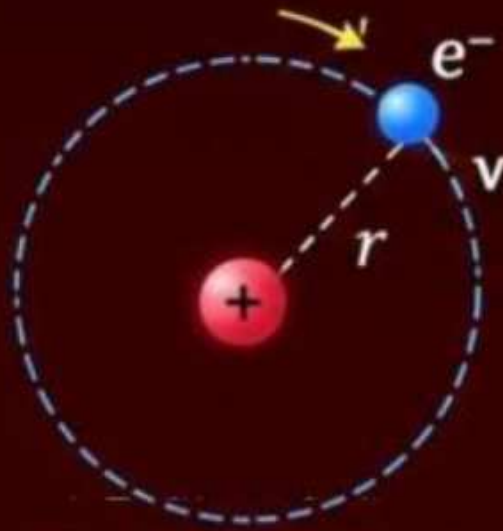


4 Angular Momentum Quantization

- Angular momentum of an electron is also **fixed (quantized)**.

$$mvr = \frac{nh}{2\pi} \quad n = 1, 2, 3, \dots$$

m = mass of electron r = radius of orbit
 v = speed h = Planck's constant



5 Ground State

- In the lowest orbit (ground state), electrons keep moving **without losing energy**.
- Energy can't be continuously lost or gained.

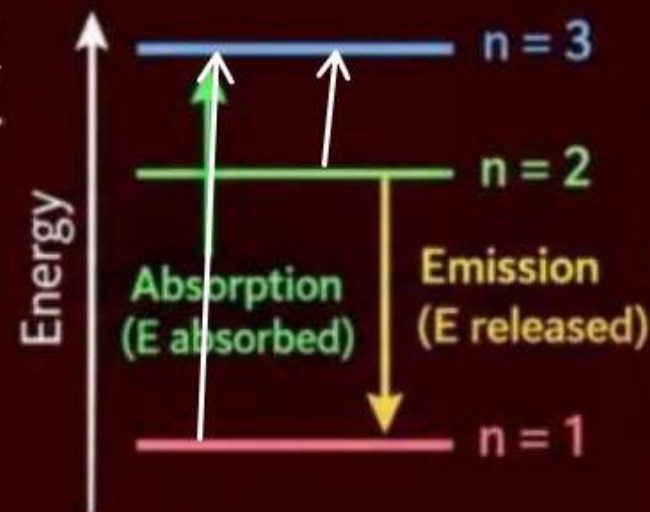


6 Energy Absorption / Emission

- Energy is **absorbed** when an electron jumps to a **higher orbit** (n increases).
- Energy is **released** when it falls to a **lower orbit** (n decreases).
- Amount of energy exchanged:

$$\Delta E = E_2 - E_1$$

Excited state
Ground state

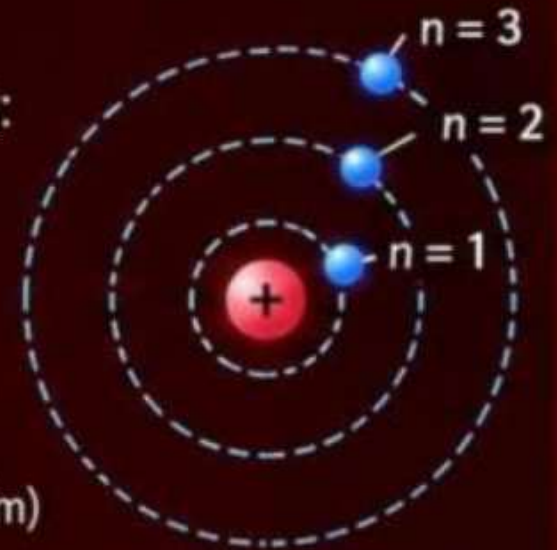


7 Radius of Orbit

- Radius of electron orbit is given by:

$$r_n = \frac{a_0 n^2}{Z}$$

r_n = radius of n th orbit
 a_0 = Bohr radius ($\approx 0.529 \text{ \AA} = 0.529 \times 10^{-10} \text{ m}$)
 Z = atomic number
 $n = 1, 2, 3, \dots$
or 52.9 pm.



radius of 2nd orbit of He^+

$$r_n = \frac{a_0 n^2}{Z} = \frac{a_0 \times 2^2}{2} = \frac{4}{2} a_0 = \boxed{2a_0}$$

Q. If energy of first is $x \text{ kJ mol}^{-1}$; the energy of 3rd orbit

a) will be also $x \text{ kJ mol}^{-1}$

b) will be more than $x \text{ kJ mol}^{-1}$

c) will be less ,, $x \text{ kJ mol}^{-1}$

d) cannot be determined

$$E_1 < E_2 < E_3 \dots$$

← 3 _____

2 _____

$x \text{ kJ mol}^{-1} \quad n=1$ _____



Limitations of Bohr's Model of Atom

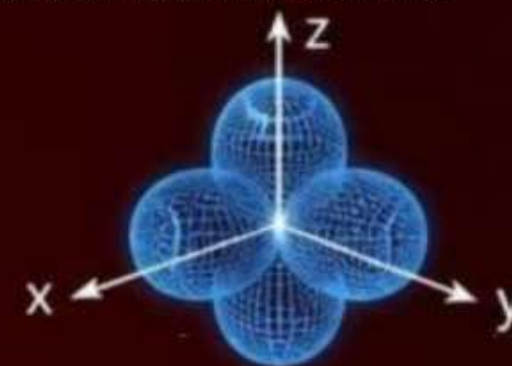
1 Inability to explain **line spectra** of multi-electron atoms.



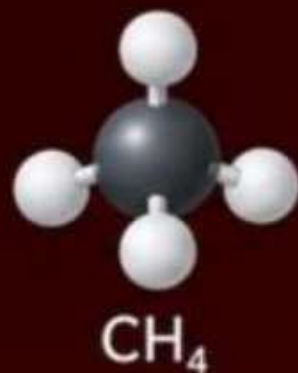
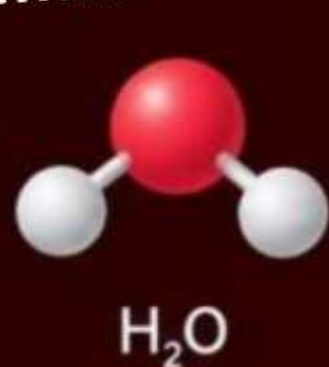
2 Inability to explain **splitting of lines** in the magnetic field (**Zeeman effect**). *+ Stark effect (EF).*



3 Inability to explain the **three dimensional** model of atom.



4 Inability to explain the **shapes of molecules**.



5 Inability to explain **de Broglie concept** of **dual character of matter** and **Heisenberg's uncertainty principle**.



Dual Nature of Matter and Radiation



1 Light shows both **wave** and **particle** nature.

- As a **wave**, it explains reflection, refraction, diffraction, etc.
 - As **particles (photons)**, Einstein explained the **photoelectric effect**.
- Black body radiation.*

This is called the **dual nature of radiation**.

Wave nature



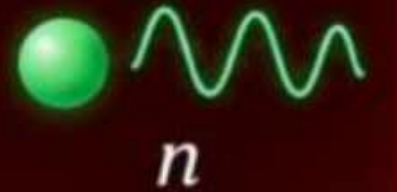
Explains reflection, refraction, diffraction, etc.

Particle nature (photons)



Explains the photoelectric effect (Einstein).

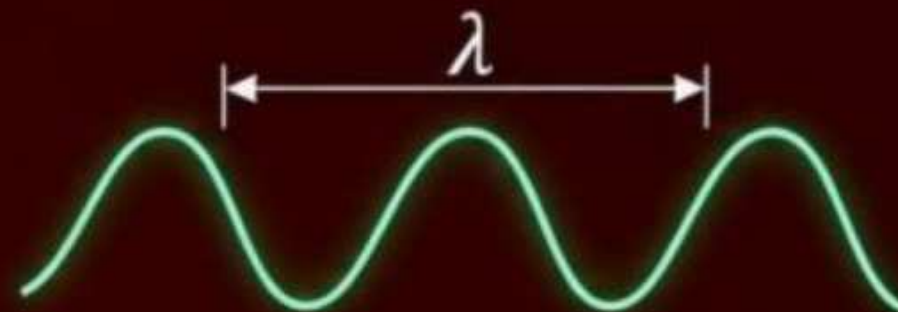
2 Later, Louis de Broglie suggested that not only light, but even microscopic particles like **electrons**, **protons**, **neutrons** also behave like waves.



3 The wavelength of such matter waves is:

$$\lambda = \frac{h}{mv}$$

where h = Planck's constant, m = mass of particle, v = velocity



Particle with matter wave

De Broglie Equation (1924)

According to Einstein's mass-energy relationship:

$$E = mc^2 \quad \text{--- (1)}$$

According to Planck's Quantum Theory:

$$E = h\nu \quad \text{--- (2) } (\nu = \text{frequency})$$

$$mc^2 = h\nu$$

$$mc^2 = h \frac{c}{\lambda} \quad \left(\because \nu = \frac{c}{\lambda} \right)$$

$$mc = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

If we replace 'c' with 'v'

$$\lambda = \frac{h}{mv}$$

or $\lambda = \frac{h}{p}$



Particle
with
matter wave

where h = Planck's constant
 m = mass of particle
 v = velocity
 p = momentum (mv)



De Broglie Equation (1924)



Significance:

- Important only for microscopic particles (electron, proton, neutron).
- For macroscopic objects, mass is very large, so wavelength becomes extremely small (nearly zero), hence negligible.

$$\lambda = \frac{h}{mv}$$

macroscopic, mass $\uparrow$, λ $\downarrow$ can't be calculate.

Heisenberg's Uncertainty Principle (1927)

- Proposed by Werner Heisenberg.
- It states that it is impossible to know simultaneously the exact position and exact momentum (or velocity) of an electron. (microscopic particle).

Mathematical form:

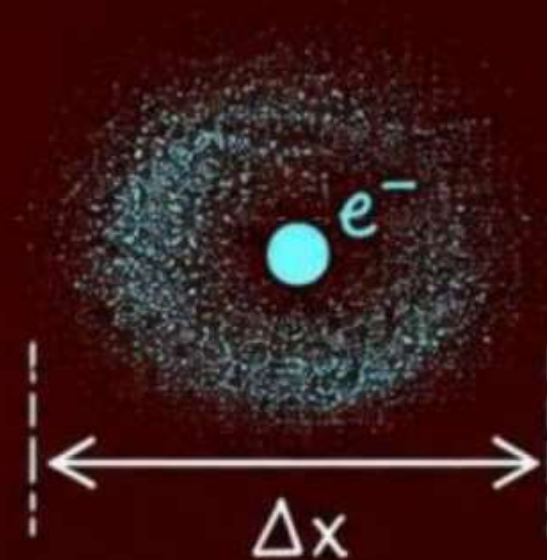
$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

or $\Delta x \cdot m\Delta v \geq \frac{h}{4\pi}$

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

Δx = Uncertainty in position

Δp = " " " momentum.



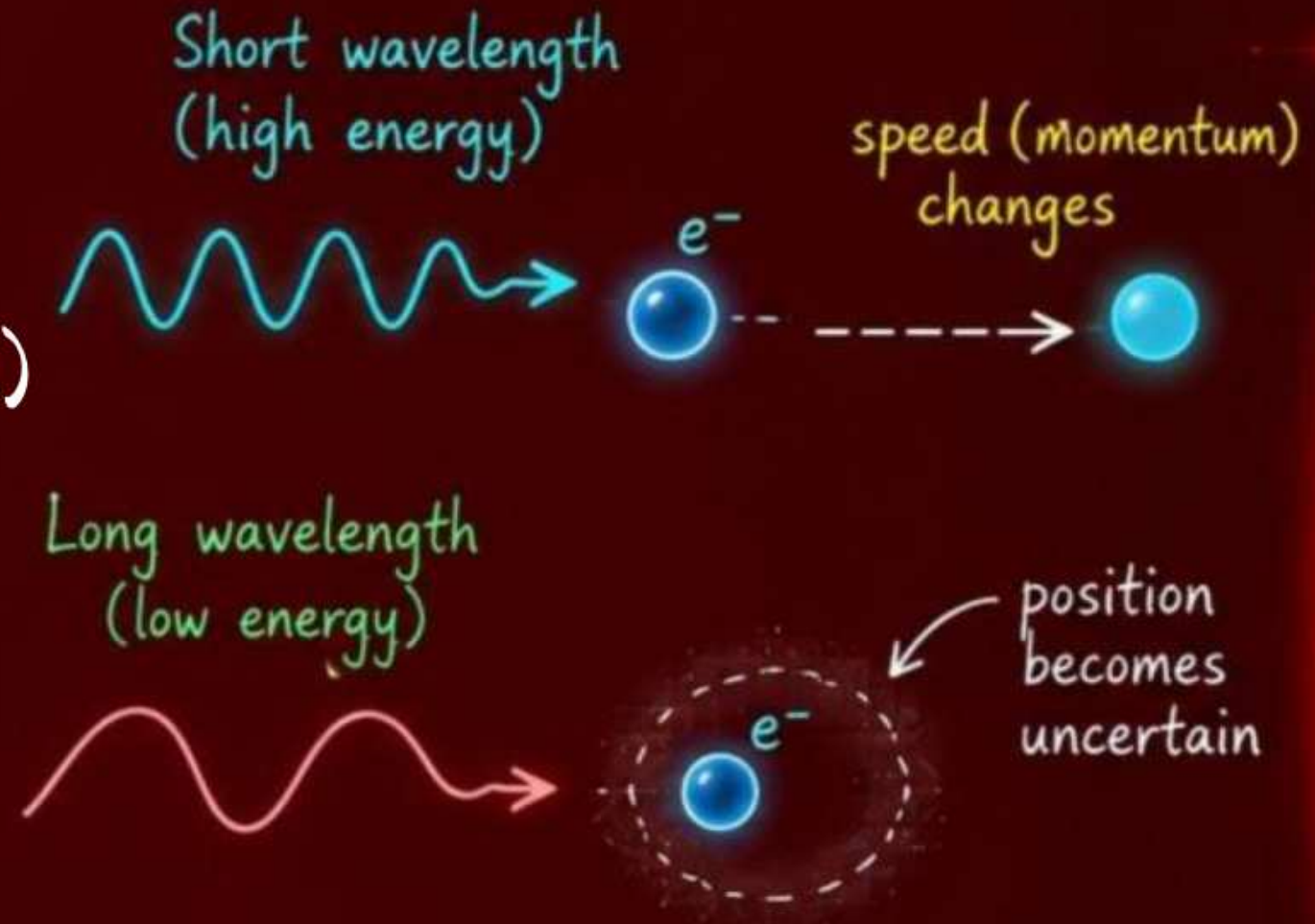
Can't know both exactly.

Heisenberg's Uncertainty Principle (1927)



Reason:

- To locate an electron exactly, we need light of very short wavelength (high energy). (Bright light)
- But when such light strikes the electron, its speed (momentum) changes.
- If we use low-energy light, its position becomes uncertain.
- Hence, both position and momentum cannot be measured together with accuracy.

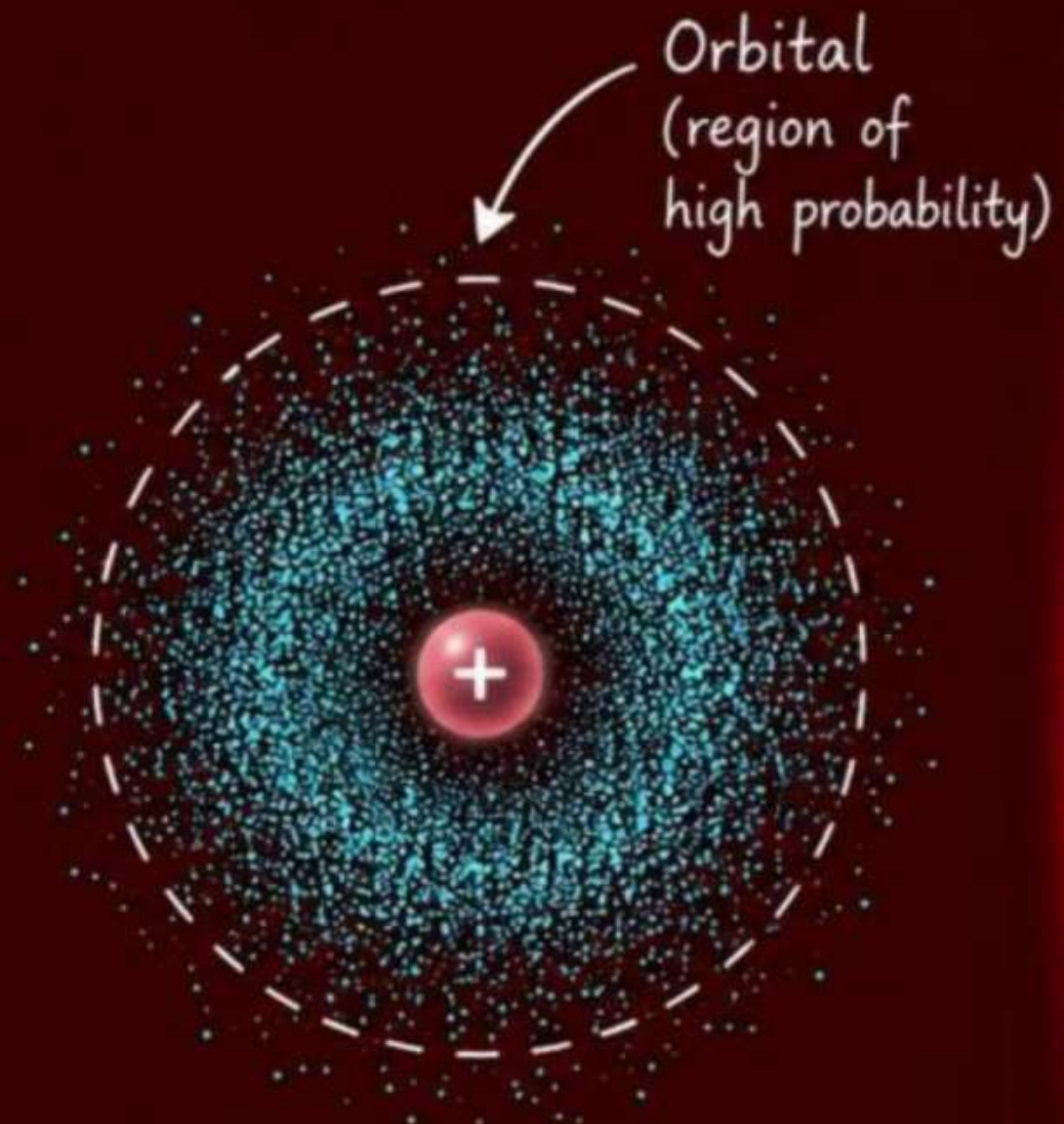


Hence, both position and momentum cannot be measured together with accuracy.

Concept of Atomic Orbitals



- The uncertainty principle showed that electrons don't move in fixed circular paths, as Bohr suggested. Instead, we can only talk about the chances of finding an electron in a certain area around the nucleus at any moment.
- " An orbital may be defined as the three dimensional space around the nucleus within which the probability of finding an electron of given energy is maximum (say up to 90%).
- Probability of finding an electron even far from the nucleus is never zero.



Probability is never zero, even far from nucleus.

8.11

Differences between Orbit and Orbital

(Key Difference)



Orbit

Orbital

1.	A well-defined circular path around the nucleus in which an electron revolves.		1.	Region in space around the nucleus where the probability of finding electron is maximum.	
2.	Represents planar (2D) motion of an electron.		2.	Represents three-dimensional (3D) motion of an electron.	
3.	Gives a definite path , which is not in accordance with the uncertainty principle.		3.	Does not specify a definite path ; electron may be anywhere in the region. This is in accordance with the uncertainty principle.	
4.	All orbits are circular .		4.	Orbitals have different shapes (e.g. s – spherical, p – dumb bell).	
5.	Do not have directional characteristics.		5.	Except s-orbitals, all other orbitals have directional characteristics.	
6.	Can accommodate $2n^2$ electrons (in the nth orbit).		6.	An orbital cannot accommodate more than two electrons .	

Quantum Numbers



Most Important



Quantum numbers may be defined as a set of four numbers with the help of which we can get complete information about all the electrons in an atom, i.e., location, energy, the type of orbital occupied, shape and orientation of that orbital, etc.

- | | | |
|---|--------------------------|-----|
| ① | Principal quantum number | P |
| ② | Azimuthal quantum number | A |
| ③ | Magnetic quantum number | M |
| ④ | Spin quantum number | S |

Quantum Numbers



1. Principal Quantum Number (n)

- Defines the **main energy level (shell)** where the electron is present.
- Denoted by $n = 1, 2, 3, \dots$
- Gives information about:

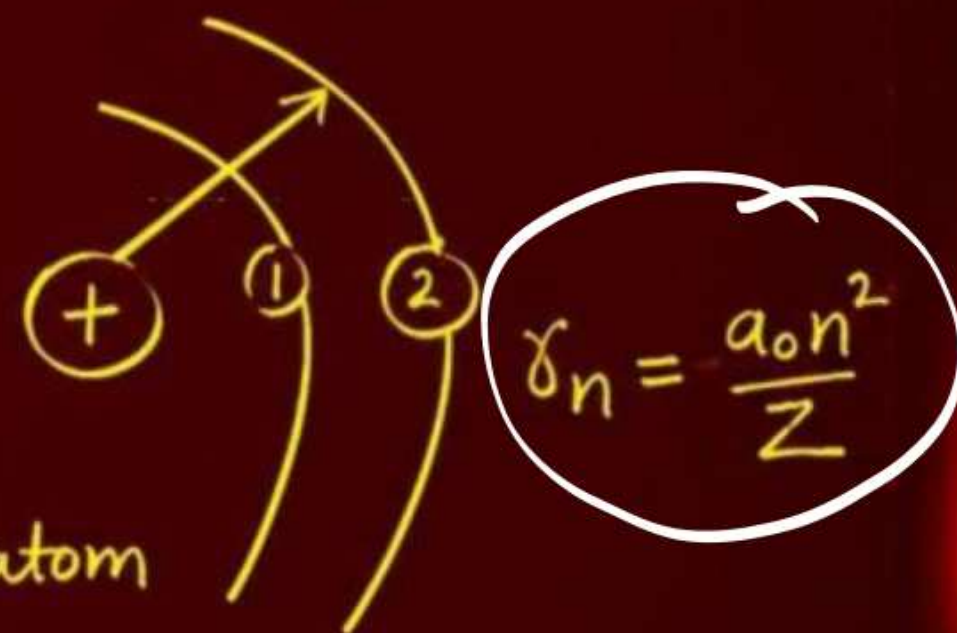
a. Average **distance** of electron from nucleus. (radius)

b. **Energy** of electron in a shell:

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV/atom}$$

$$\therefore \text{No. of } e^- \text{ in a shell} = \underline{2n^2}$$

$$5^{\text{th}} \text{ shell} = 2 \times 5^2 = 2 \times 25 = 50$$



Quantum Numbers



$$\begin{matrix} p \rightarrow n \\ A \rightarrow l \end{matrix} \quad (n = 0 \text{ to } (n-1))$$



2. Azimuthal Quantum Number (l)

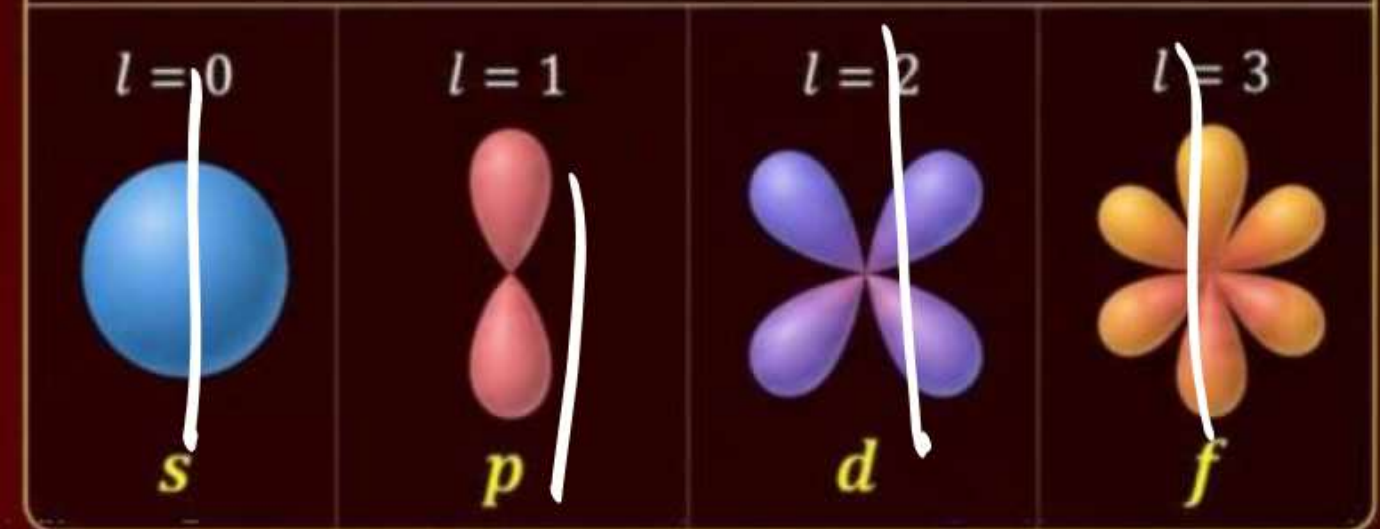
- Also called **subsidiary quantum number**.
- Defines the **subshells** inside a principal shell.
- For any value of n ,

$$l = 0 \text{ to } (n - 1)$$

Examples: $l = 0 + 1(1-1) = 0$

- $n = 1 \rightarrow l = 0 \rightarrow 1$ subshell (s)
- $n = 2 \rightarrow l = 0, 1 \rightarrow 2$ subshells (s, p)
- $n = 3 \rightarrow l = 0, 1, 2 \rightarrow 3$ subshells (s, p, d)
- $n = 4 \rightarrow l = 0, 1, 2, 3 \rightarrow 4$ subshells (s, p, d, f)

Subshells and their symbols



n	Possible values of l	No. of subshells	Subshells
1	0	1	s
2	0, 1	2	s, p
3	0, 1, 2	3	s, p, d
4	0, 1, 2, 3	4	s, p, d, f

Quantum Numbers

$$\begin{aligned} p &= n \\ A &= l \\ M &= m_l \end{aligned}$$



3. Magnetic Quantum Number (m_l)

- Defines the **number of orbitals** in a given subshell.
- For a given l , values of m_l range from $-l$ to $+l$ (including 0).

Examples:

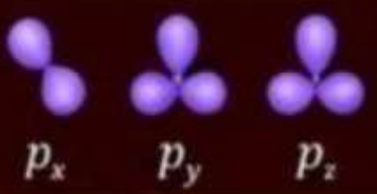
- $l = 0$ (s subshell) $m_l = 0 \rightarrow 1$ orbital
- $l = 1$ (p subshell) $m_l = -1, 0, +1 \rightarrow 3$ orbitals
- $l = 2$ (d subshell) $m_l = -2, -1, 0, +1, +2 \rightarrow 5$ orbitals
- $l = 3$ (f subshell) $m_l = -3$ to $+3 \rightarrow 7$ orbitals

$\therefore$ Number of orbitals in a subshell = $2l + 1$ • n^2 (for a shell)

No. of orbitals in a shell

For a given l ,
 $m_l = -l, -l + 1, \dots, 0, \dots, +l$
 Total orbitals = $2l + 1$

Subshell, m_l values and number of orbitals

l	Subshell	m_l values	Number of orbitals	Pictorial representation
0	s	0	1	 s
1	p	$-1, 0, +1$	3	 p_x p_y p_z
2	d	$-2, -1, 0, +1, +2$	5	 d
3	f	$-3, -2, -1, 0, +1, +2, +3$	7	 f

Shell (n)



Subshell (l)



Orbital (m_l)



Quantum Numbers



4. Spin Quantum Number (m_s)

- Defines the **orientation of spin** of an electron in an orbital.
- An electron can have **two possible spins**:

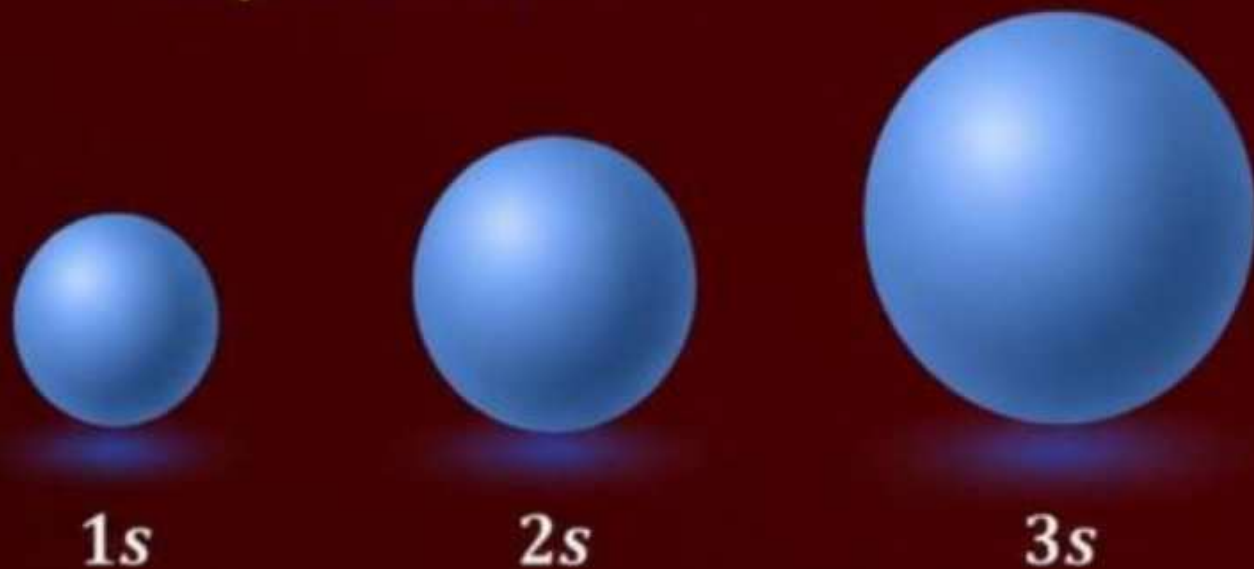
- **Clockwise** $\rightarrow m_s = +\frac{1}{2}$ ($\uparrow$)   $m_s = +\frac{1}{2}$

- **Anticlockwise** $\rightarrow m_s = -\frac{1}{2}$ ($\downarrow$)   $m_s = -\frac{1}{2}$

Shapes of Orbitals

(a) Shape of s-orbital

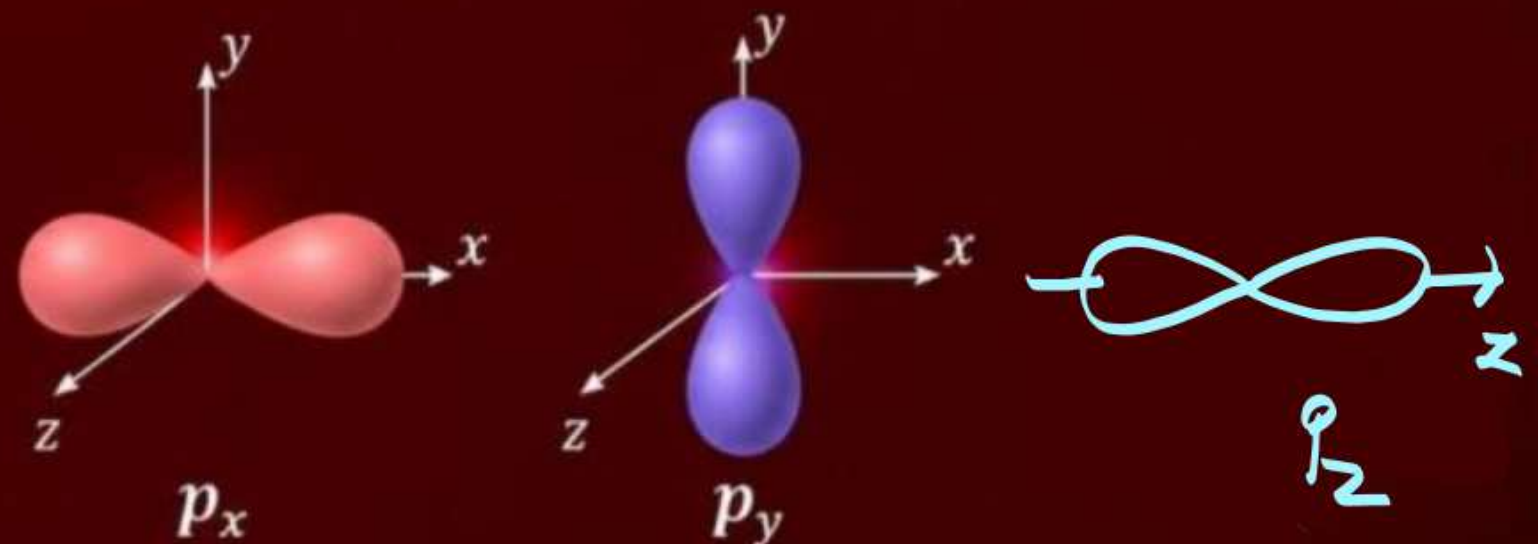
- According to probability calculations of quantum mechanics, *s*-orbital has **spherical shape**.
- For $l = 0$ (*s*), $m_l = 0$ (only one value) → **only 1 orbital**.



***s*-orbitals are spherical.**

(b) Shape of p-orbitals

- According to probability calculations of quantum mechanics, *p*-orbitals have **dumbbell shape**.
- For $l = 1$ (*p*), $m_l = -1, 0, +1$ → **3 orbitals** (p_x, p_y, p_z).



***p*-orbitals are dumbbell shaped.**

Shapes of Orbitals

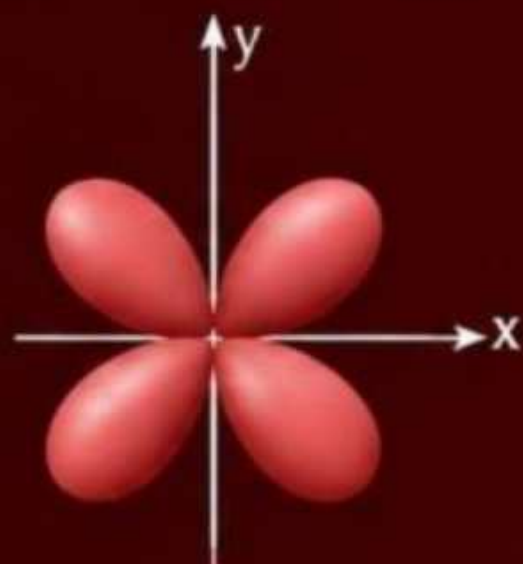


(c) Shape of d-orbitals

- d-orbitals have **double dumbbell** or **clover leaf shape**.
- For $l = 2$ (d), $m_l = -2, -1, 0, +1, +2$ (5 orbitals).

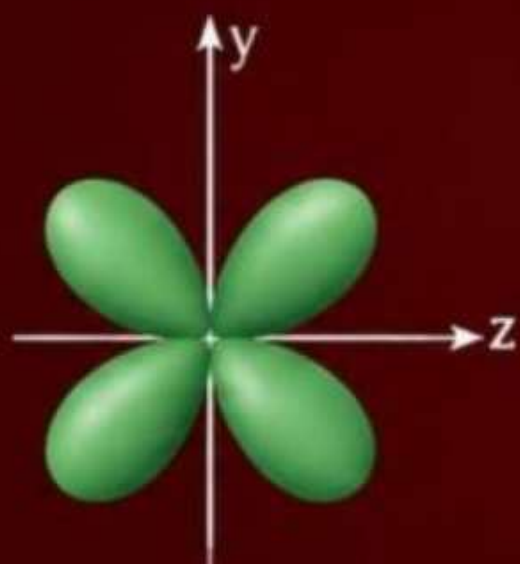
$$\text{Number of d-orbitals} = 2l + 1 = 5$$

(for $l = 2$)



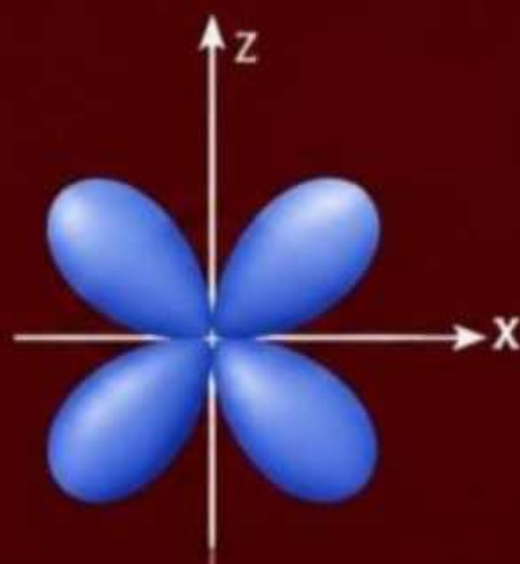
d_{xy}

Lobes lie between x and y axes.



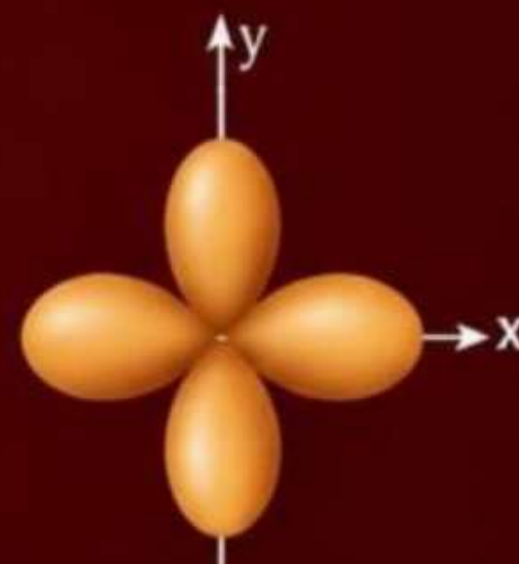
d_{yz}

Lobes lie between y and z axes.



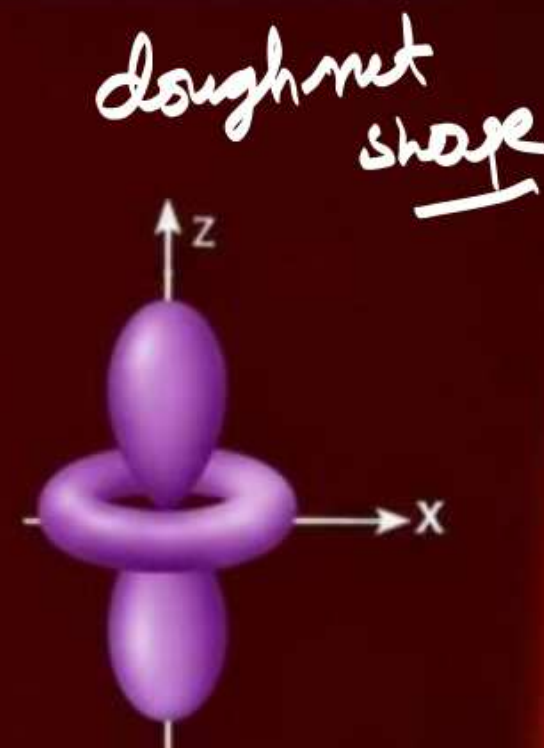
d_{xz}

Lobes lie between x and z axes.



$d_{x^2-y^2}$

Lobes along the x and y axes.



d_{z^2}

Two lobes along z axis with a toroidal ring.

d-orbitals are double dumbbell or clover leaf shaped.

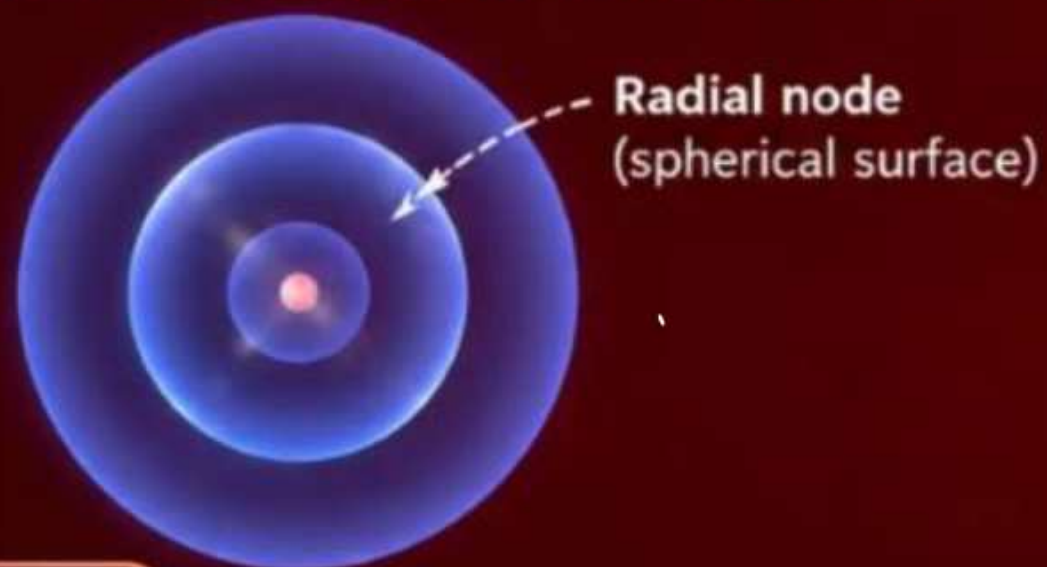
Non-Axial

Axial

doughnut shape

1. Radial Node (Spherical node)

- A **spherical region** around the nucleus where the probability of finding an electron is **zero**.

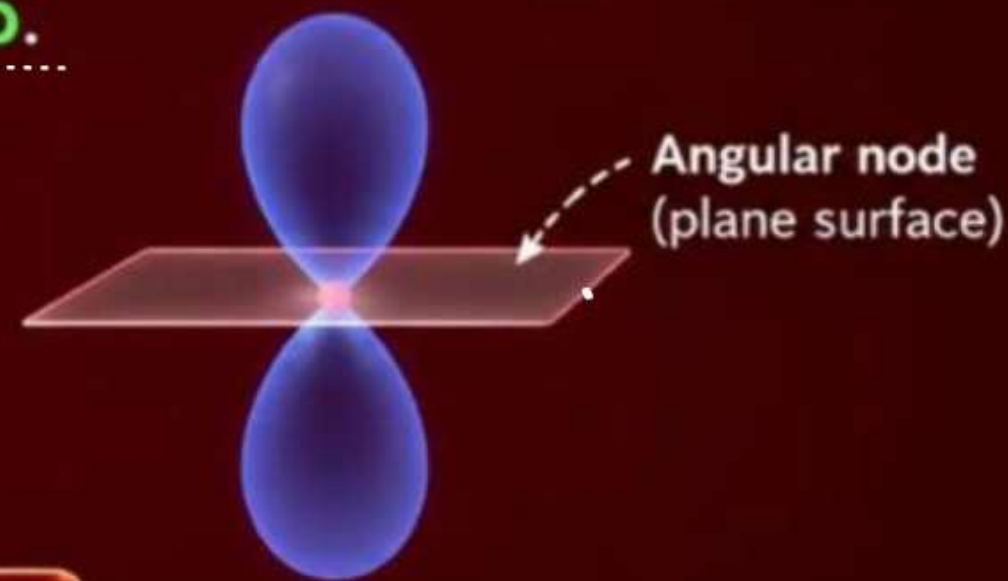


Formula:

$$\text{Radial nodes} = \underline{n} - \underline{l} - \underline{1}$$

2. Angular Node

- A **plane/conical surface** passing through the nucleus where the probability of finding an electron is **zero**.



Formula:

$$\text{Angular nodes} = l$$

Total Nodes: | Total nodes = $n - 1$

Q. Calculate the node in 3d

$$d, l=2$$

a) Radial $\Rightarrow (\underline{n} - l - 1) = 3 - 2 - 1 = 3 - 3 = \textcircled{0}$

b) Angular $\Rightarrow \underline{l} = \textcircled{2}$

$$\begin{aligned} \text{Total node} &= n - 1 \\ &= 3 - 1 = \textcircled{2} \end{aligned}$$

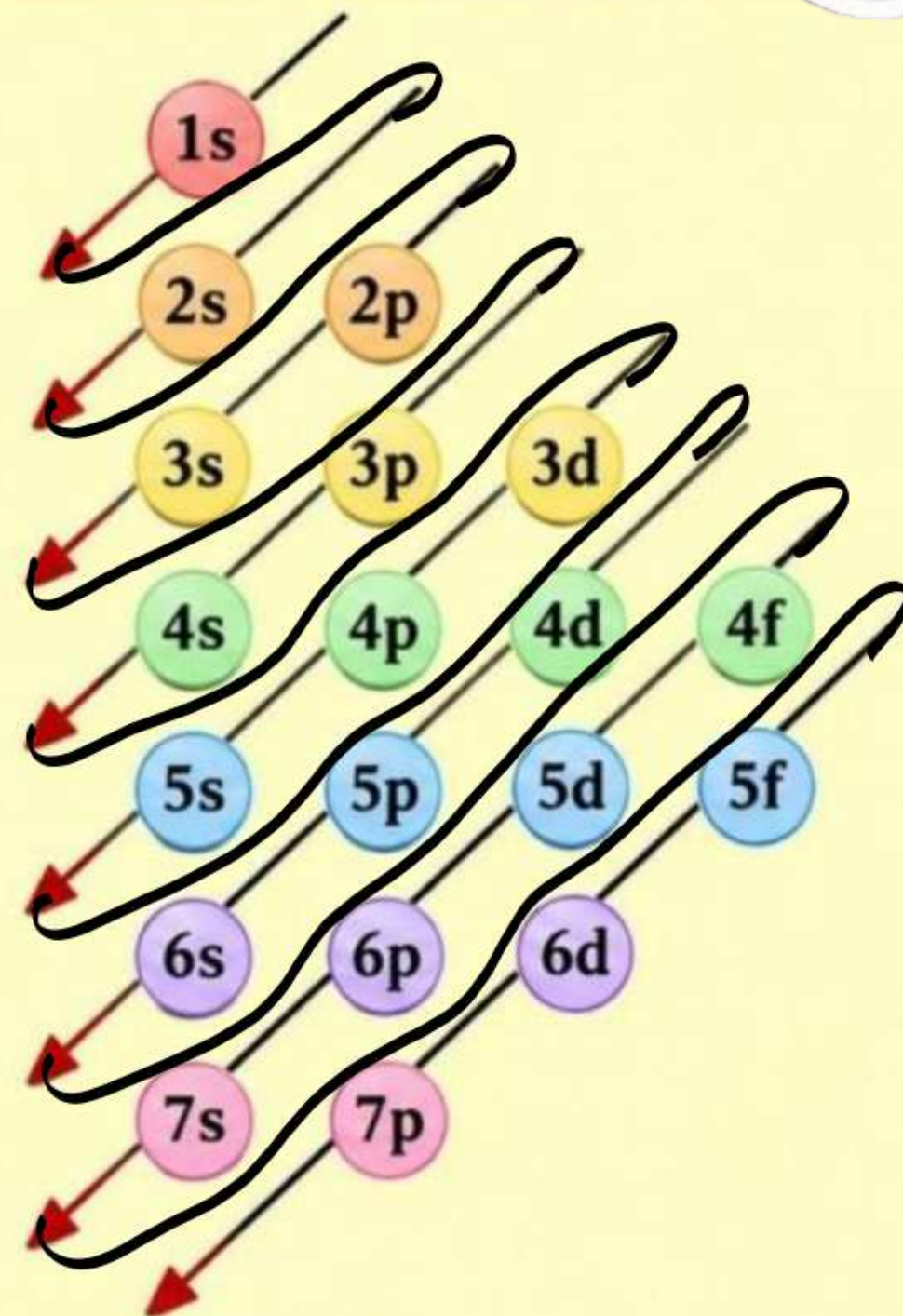
RULES OF FILLING ORBITALS

- Aufbau principle:**

“Electrons first occupy the **lowest-energy** orbital available to them and enter into **higher energy** orbitals only after the lower energy orbitals are filled.”

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p$$

Increasing
energy





RULES OF FILLING ORBITALS

- Energy of an orbital is determined by a rule known as the $(n + l)$ rule or **Bohr-Bury's rule**.

According to this rule:

- Higher $(n + l)$ value $\rightarrow$ Higher energy
- If $(n + l)$ is same $\rightarrow$ Lower n value has lower energy

$(n + l)$ values

$$s \ (l = 0) \rightarrow n + 0 = n$$

$$p \ (l = 1) \rightarrow n + 1$$

$$d \ (l = 2) \rightarrow n + 2$$

$$f \ (l = 3) \rightarrow n + 3$$

For the same n :

$$s < p < d < f \text{ (in energy)}$$

1 Higher $(n + l) \rightarrow$ Higher energy

- $4s: n + l = 4 + 0 = 4$
- $3d: n + l = 3 + 2 = 5$

$$\therefore 3d > 4s \text{ in energy}$$

2 If $(n + l)$ is same $\rightarrow$ Lower $n =$ Lower energy

- $2p: n + l = 2 + 1 = 3$
- $3s: n + l = 3 + 0 = 3$

$$\therefore 2p < 3s \text{ in energy}$$

$$(n+l) \ 2p = 2+1 = 3$$
$$3s = 3+0 = \textcircled{3}$$

Orbital with smaller $(n + l)$ has lower energy.
If $(n + l)$ is same, orbital with lower n has lower energy.



$$3s < 3p < \boxed{4s < 3d}$$

$$n+l(4s) = 4+0 = \textcircled{4}$$

$$3d = 3+2 = \textcircled{5}$$

RULES OF FILLING ORBITALS

Pauli's exclusion principle

- According to this principle, an orbital can accommodate maximum of two electrons and these must have opposite spins.

OR

No two electrons in an atom can have the **same set of four quantum numbers**.

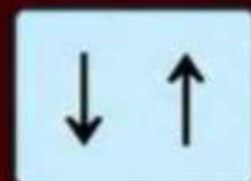
Key Points:

- Maximum **2 electrons** per orbital.
- Spins must be **opposite** ($+\frac{1}{2}$ and $-\frac{1}{2}$).
- If **n , l and m** are same for two electrons, then their **spin (m_s)** must be **different**.

Correct representations:

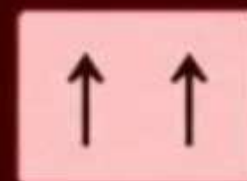


(i)

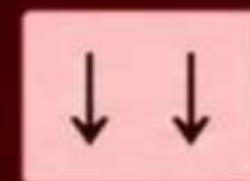


(ii)

Incorrect representations:



(iii)



(iv)

If for two electrons the three quantum numbers **n , l and m** have the **same value**, the fourth quantum number (**spin, m_s**) is different i.e. $+\frac{1}{2}$ for one electron and $-\frac{1}{2}$ for the other.

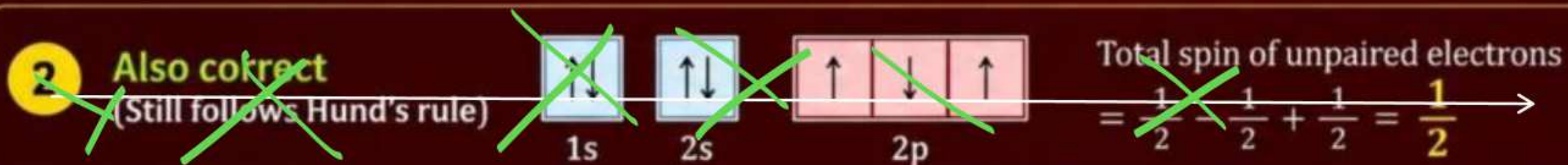
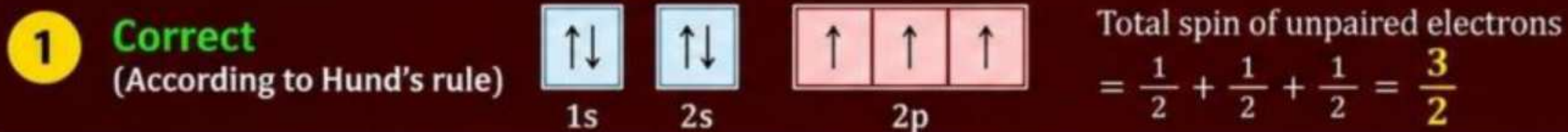
RULES OF FILLING ORBITALS



Hund's Rule of Maximum Multiplicity

According to this rule, pairing of electrons in the orbitals belonging to the same subshell (p , d or f) i.e. degenerate orbitals, **does not take place** until **each orbital** belonging to that subshell has got **one electron each**, i.e. it is **singly occupied**.

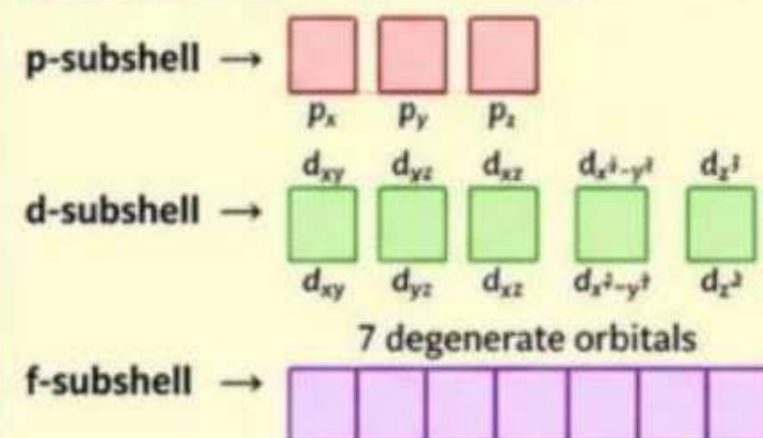
Example: Electronic configuration of nitrogen ($Z = 7$): $1s^2 2s^2 2p^3$



Key Points:

- Applies to degenerate orbitals (same subshell: p , d or f).
- Electrons occupy all orbitals singly with **parallel spins** first.
- Pairing starts only after each orbital has one electron.

Degenerate orbitals:



Electrons in degenerate orbitals occupy all orbitals singly with parallel spins first.
Pairing starts only after each orbital has one electron.

Electron Configuration

- ① H = $1s^1$
- ② He = $1s^2$
- ③ Li = $[He] 2s^1$ or $1s^2 2s^1$
- ④ Be = $[He] 2s^2$
- ⑤ B = $[He] 2s^2 2p^1$
- ⑥ C = $[He] 2s^2 2p^2$ or $1s^2 2s^2 2p^2$
- ⑦ N = $[He] 2s^2 2p^3$
- ⑧ O = $[He] 2s^2 2p^4$
- ⑨ F = $[He] 2s^2 2p^5$
- ⑩ Ne = $[He] 2s^2 2p^6$

Order of energy

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p$$

Subshell capacity

$$s^2, p^6, d^{10}, f^{14}$$

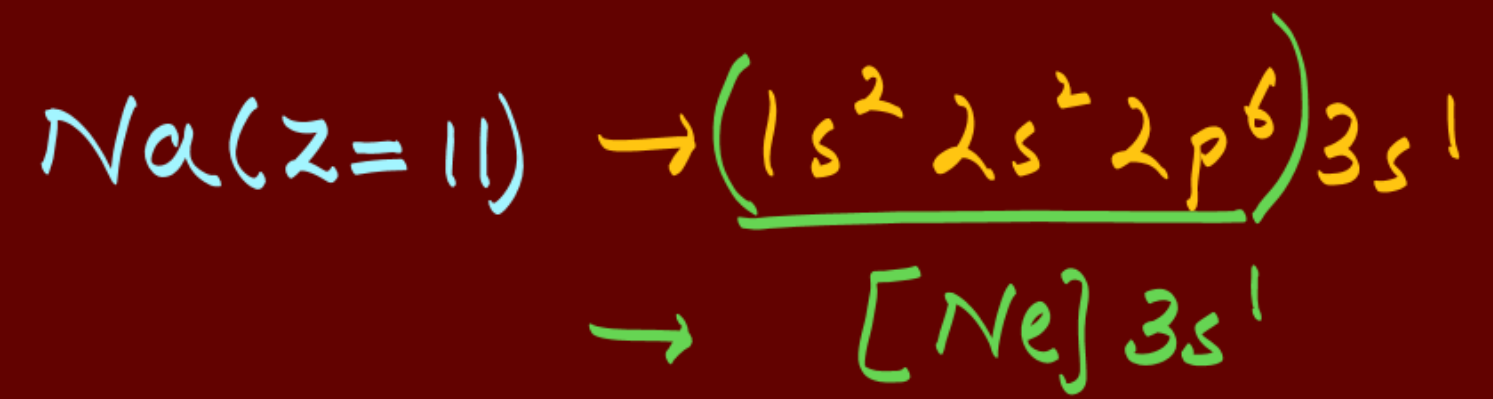
$$s - \text{subshell} = 2e^-$$

$$p - \text{subshell} = 6e^- \left(\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} \right)$$

$$d - \text{subshell} = 10e^- \left(\begin{array}{|c|c|c|c|c|} \hline & & & & \\ \hline \end{array} \right)$$

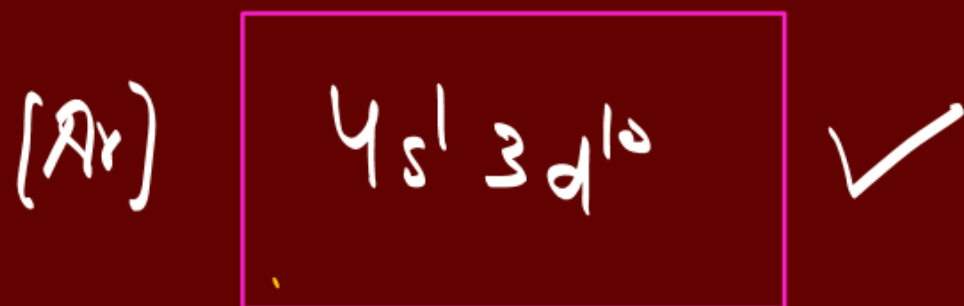
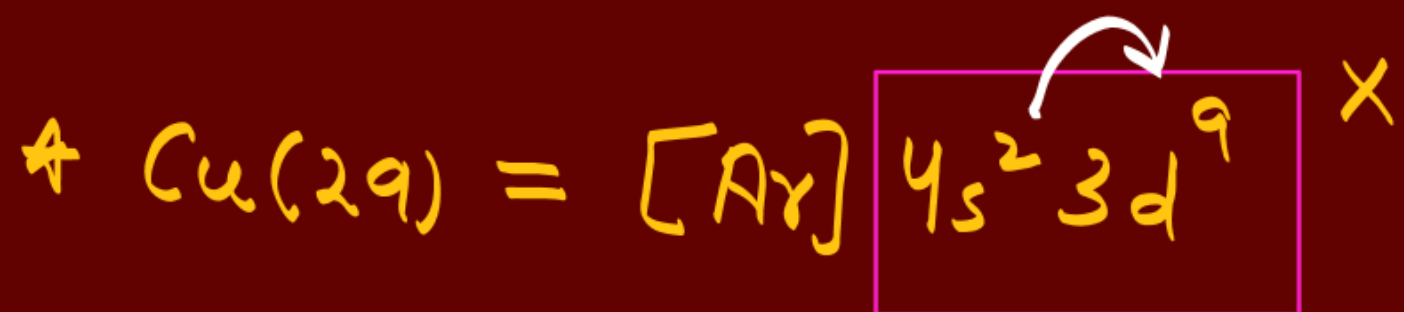
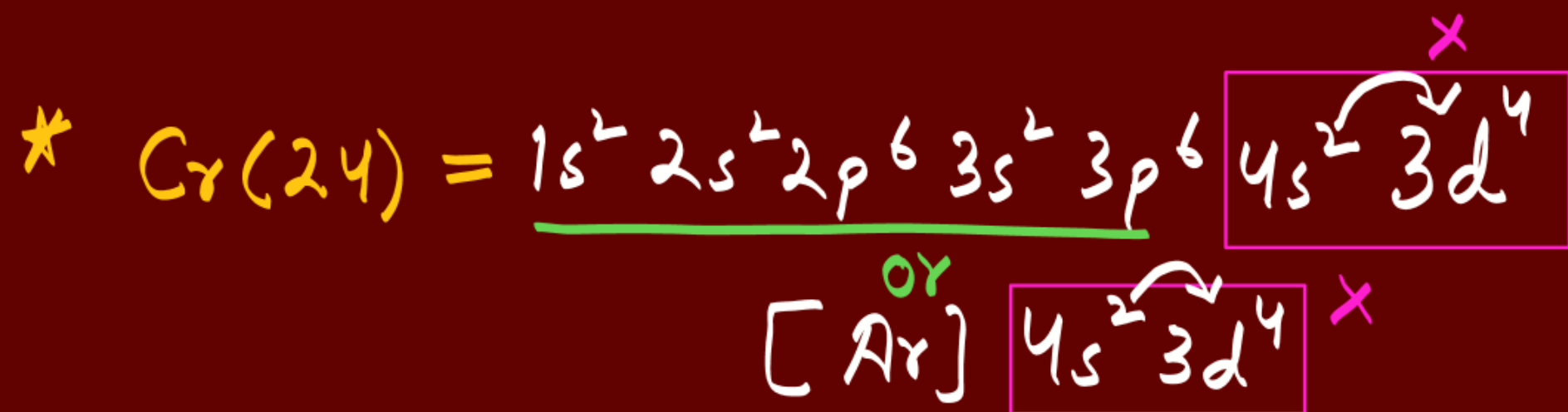
$$f - \text{subshell} = 14e^- \left(\begin{array}{|c|c|c|c|c|c|c|} \hline & & & & & & \\ \hline \end{array} \right)$$

He	(Z = 2)	2	+2
Ne	(Z = 10)	2	+8
Ar	(Z = 18)	2	+16
Kr	(Z = 36)	2	+34
Xe	(Z = 54)		
Rn	(Z = 86)		

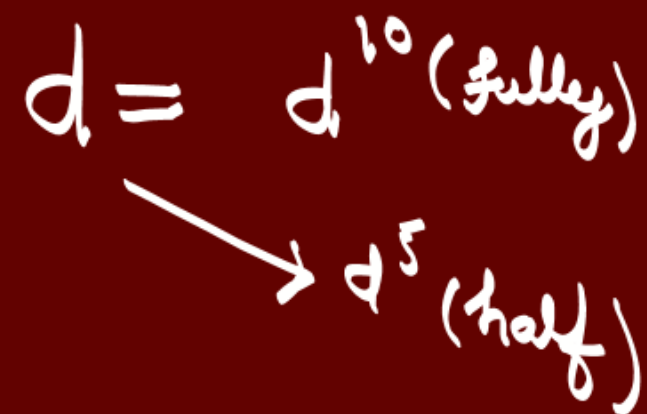


1 to 20

20 to 30



Exceptional





ELECTRONIC CONFIGURATION OF IONS $\rightarrow$

e.g Fe^{2+} ($Z=26$)

Sol: $\rightarrow \text{Fe} (Z=26) = [\text{Ar}] 4s^2 3d^6$

Fe^{2+} ($Z=26$) = $[\text{Ar}] \underline{4s^0} 3d^6$ ✓

Cations K case
Me $\bar{e}$ remove
Kste has highest
'n'wala orbital
sq''

Electronic Configuration (Z = 1 – 30)



Z	Element	Electronic Configuration
1	H	$1s^1$
2	He	$1s^2$
3	Li	$1s^2 2s^1$
4	Be	$1s^2 2s^2$
5	B	$1s^2 2s^2 2p^1$
6	C	$1s^2 2s^2 2p^2$
7	N	$1s^2 2s^2 2p^3$
8	O	$1s^2 2s^2 2p^4$
9	F	$1s^2 2s^2 2p^5$
10	Ne	$1s^2 2s^2 2p^6$
11	Na	$1s^2 2s^2 2p^6 3s^1$
12	Mg	$1s^2 2s^2 2p^6 3s^2$
13	Al	$1s^2 2s^2 2p^6 3s^2 3p^1$
14	Si	$1s^2 2s^2 2p^6 3s^2 3p^2$
15	P	$1s^2 2s^2 2p^6 3s^2 3p^3$

Z	Element	Electronic Configuration
16	S	$1s^2 2s^2 2p^6 3s^2 3p^4$
17	Cl	$1s^2 2s^2 2p^6 3s^2 3p^5$
18	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$
19	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
20	Ca	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
21	Sc	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
22	Ti	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
23	V	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
24	Cr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
25	Mn	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
26	Fe	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
27	Co	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
28	Ni	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
29	Cu	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
30	Zn	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$

Stability of Completely Filled and Half-Filled Subshells

In some elements like **chromium (Cr)** and **copper (Cu)**, one electron **shifts from 4s to 3d** to attain more stable (half-filled or fully filled) configurations.

Completely filled (d^{10}) or half-filled (d^5) subshells are more stable than partially filled subshells.

Examples:

Cr ($Z = 24$): $[\text{Ar}] 3d^5 4s^1$
(instead of $[\text{Ar}] 3d^4 4s^2$)

Cu ($Z = 29$): $[\text{Ar}] 3d^{10} 4s^1$
(instead of $[\text{Ar}] 3d^9 4s^2$)

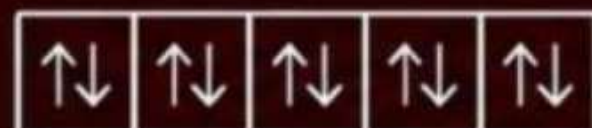
Reasons for this stability:

1 Symmetrical distribution

- Half-filled or completely filled subshells have a **symmetrical** electron arrangement.
- This **reduces electron-electron repulsion**, making them more stable.



d^5 (half-filled)



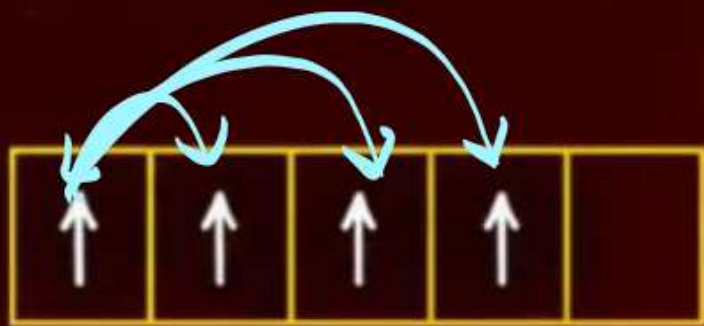
d^{10} (completely filled)

2 Exchange energy

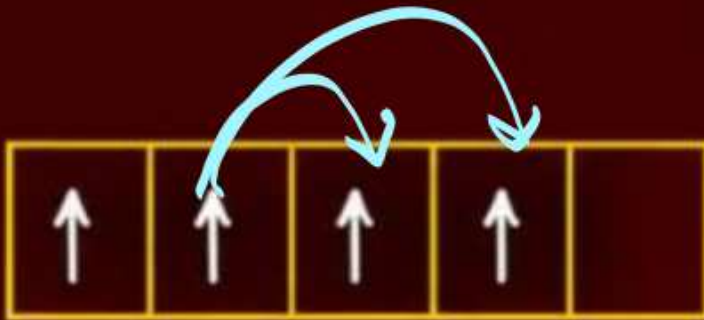
- Electrons with the **same spin** in degenerate orbitals (e.g. 3d) can exchange places.
- These exchanges **release energy** (exchange energy).
- Number of exchanges is **maximum** in half-filled or completely filled subshells → greater stability.

More exchange energy → Greater stability

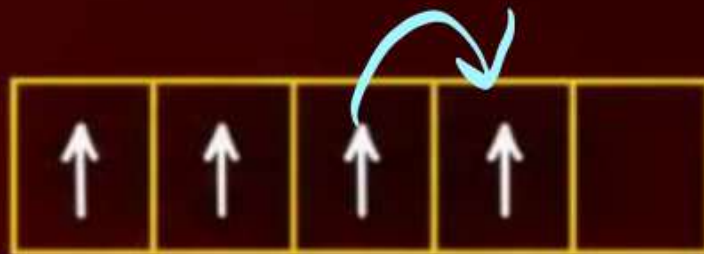
$d^4 =$



3



2

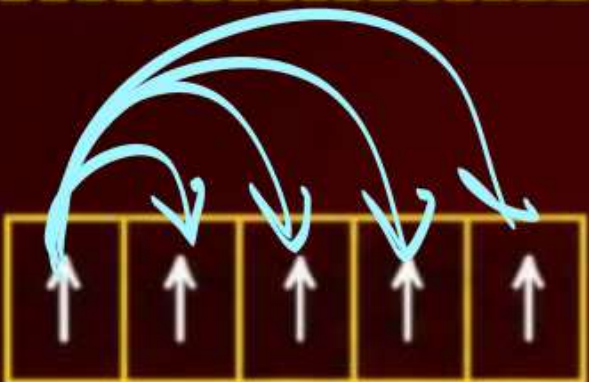


1

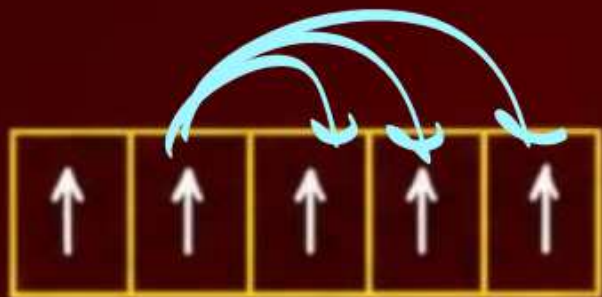
Total = 6 exchange

No. of exchange = $\frac{n(n-1)}{2}$ $n = \text{no. of } e^- \text{ of same spin}$

$d^5 =$



4



3



2



1

Total = 10 exchange

Exchange energy is maximum when number of unpaired electrons is maximum.

S-subshell $\rightarrow$ one orbital $\rightarrow$ $\uparrow\downarrow$
s

p-subshell $\rightarrow$

($l=1$)

$m_l = +1, 0, -1$ (3-orbitals)

$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$
 p_x p_y p_z

QUESTION



What are Quantum Numbers ? Explain the significance of any two of the Quantum numbers. [2025]

3 marks

QUESTION



The electronic configuration of Fe^{2+} is:

[2024]

- 1 $[\text{Ar}] 4s^2 3d^4$
- 2 $[\text{Ar}] 4s^0 3d^6$
- 3 $[\text{Ar}] 4s^1 3d^5$
- 4 $[\text{Ar}] 4s^2 3d^6$

QUESTION



What are Quantum Numbers? Briefly explain the significance of each quantum number .

OR

Write the main postulates of Bohr's theory and what are its limitations?

How this model helped in calculating the energy of an electron in different energy levels of hydrogen atom? [2024]

QUESTION



Define an atomic orbital. Differentiate between orbit and orbital.

[2024]

QUESTION



Number of orbitals in a shell having principle Quantum number n is:

1 $n + 2$

2 $2n^2$

3 n^2

4 $2n$

QUESTION



The Principle Quantum Number of an electron represents:

[2025]

- 1 Size of the orbital
- 2 Spin angular momentum
- 3 Orbital angular momentum
- 4 Space orientation of the e^-

QUESTION



The discovery of nucleus was done by:

[2025]

- 1 J. J. Thomson
- 2 Chadwick
- 3 Rutherford
- 4 Moseley

QUESTION



State and explain Heisenberg's uncertainty principle.

[2025]

OR

Define an atomic orbital. Discuss the shapes of s , p and d orbitals.



Zoom SHAKA LAKA.

Thank
You